


# Mr. Mars Music Studio All in 1

User Manual

DGN.V.06

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# 1. Getting started

Mr. Mars Music Studio All in 1 is a music production environment that runs in your browser. It combines multitrack recording, MIDI editing, a mixer, guitar and bass amp simulation, a DJ mixer, a tool that turns audio into notes, and a plugin system. It needs no installation, and once started it works without internet.

## What you need

- **Computer:** Windows or Mac with Microsoft Edge or Google Chrome.
- **For guitar or bass:** an audio interface with an instrument (Hi-Z) input.
- **For MIDI keyboards and DJ controllers:** Edge or Chrome. Safari has no MIDI.

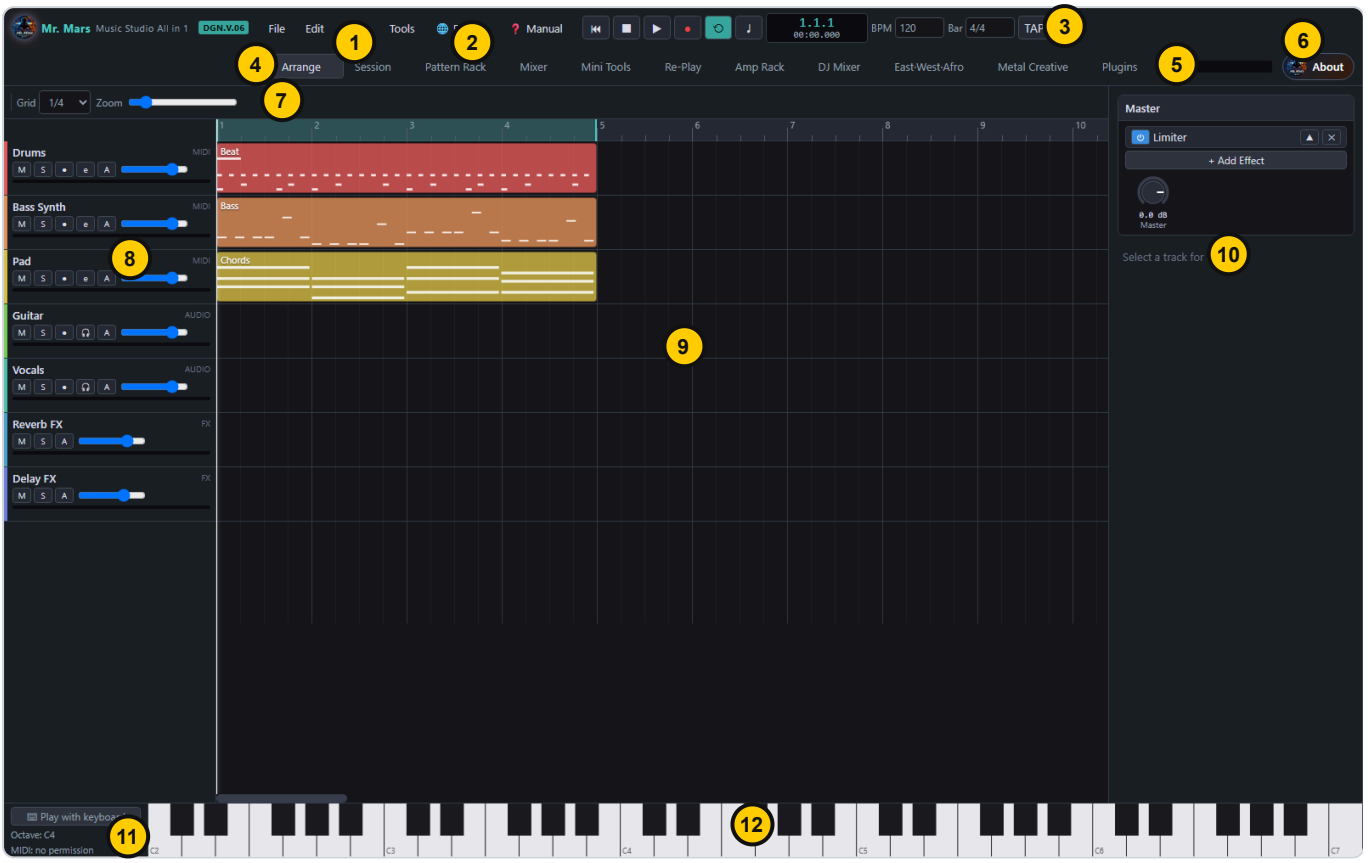
## Starting the app

| Device | How to start |
|---|---|
| Windows | Double-click `Mr Mars Music Studio.cmd`. The app opens in its own window with its own browser profile. Run `Masaustu Kisayolu Olustur.cmd` once to add a desktop and Start menu shortcut. |
| Mac | Double-click `Mr Mars Music Studio.command`. If macOS blocks it the first time, right-click it and choose Open. See chapter 17. |
| Phone or tablet | Open the app's web address, then add it to the home screen. See chapter 17. |

## Your first five minutes

1. Open **File** > **Demo project** to load a short example with drums, bass and chords.
2. Press `Space` to play and stop. The ↻ button repeats the marked loop region.
3. Double-click a coloured clip to open it in the piano roll, and change a few notes.
4. Open the **Mixer** tab and balance the tracks with the faders.
5. Press `Ctrl` + `S` to save, then use **File** > **Mixdown / Export WAV** to make a WAV file.

# 2. The main window

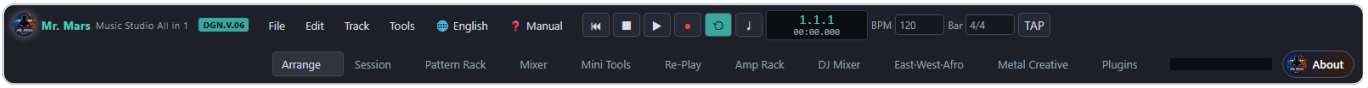



The main window with the demo project loaded.

| # | Area | What it does |
|---|---|---|
| 1 | Menus | File, Edit, Track and Tools. See chapter 3. |
| 2 | 🌐 Language, ❓ Manual | Shows the current language and lets you change it. ❓ Manual opens this manual. |
| 3 | Transport | Play, stop, record, loop, metronome, position, tempo and time signature. |
| 4 | Tabs | The eleven work areas of the app. |
| 5 | Master meter | Shows the level of the final output. |
| 6 | About | Version, audio engine, plugin count, language and data folder. |
| 7 | Tab toolbar | Tools for the current tab. It changes with each tab. |
| 8 | Track headers | Track name and its buttons and volume. |
| 9 | Timeline | Clips arranged over time, with the ruler above. |
| 10 | Inspector | Details of the selected track, or the Master when no track is selected. |
| 11 | Keyboard panel | "Play with keyboard", current octave and MIDI status. |

| # | Area | What it does |
|---|---|---|
| 12 | On-screen piano | Click the keys to play the selected instrument. |

## Transport buttons

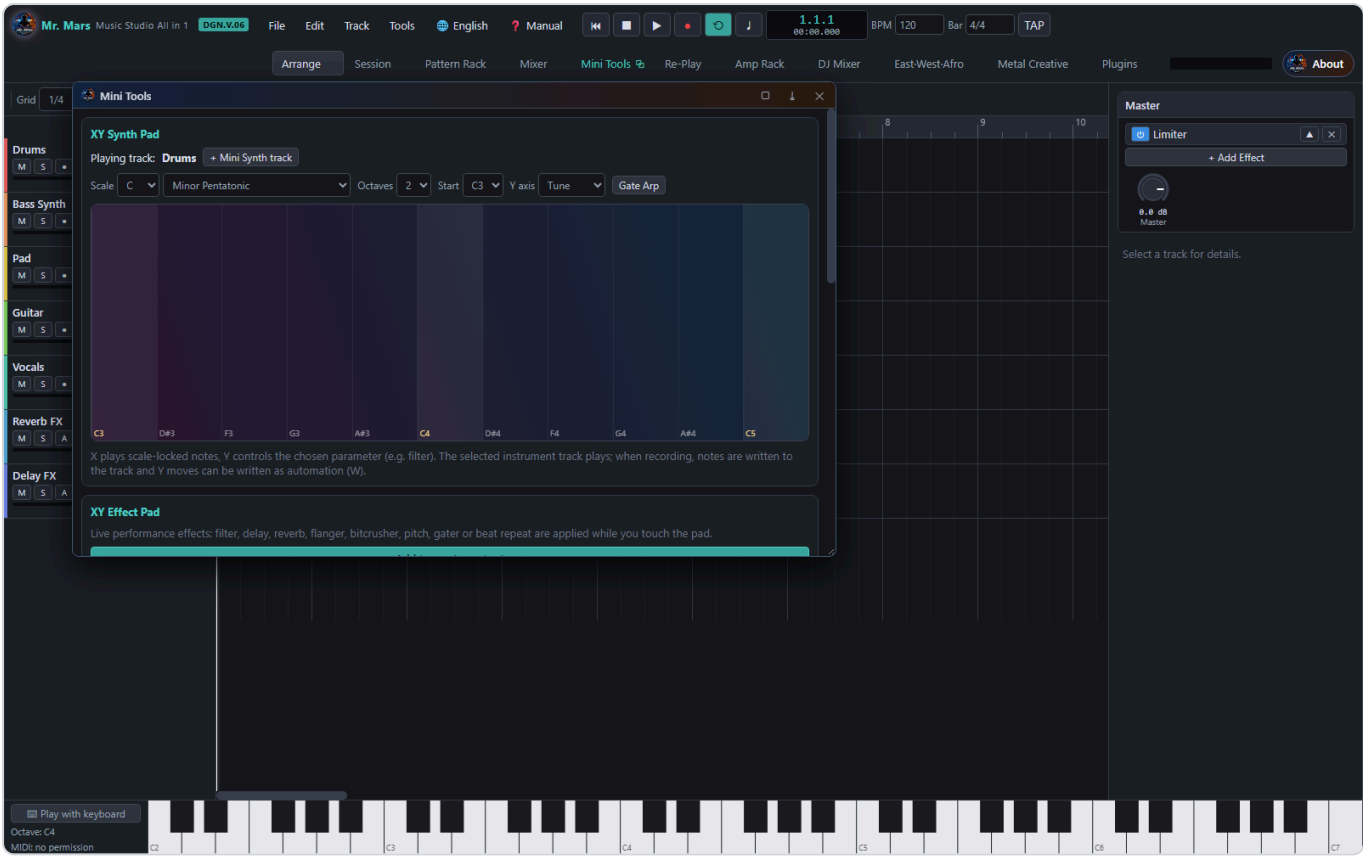



The top bar: menus, language and manual buttons, transport, tabs and About.

| Button | Function | Key |
|---|---|---|
| ⏮ | Back to the start | `Home` |
| ■ | Stop | |
| ▶ | Play or pause | `Space` |
| ● | Record on armed tracks | `R` |
| ⟲ | Loop on or off | `L` |
| ♩ | Metronome on or off | `M` |
| Display | Position in bars.beats.ticks, and in time | |
| BPM | Project tempo. Type a value from 20 to 300. | |
| Bar | Time signature, for example 4/4 | |
| TAP | Click in time to set the tempo | |

## Separate windows

Any tab can open in its own floating window, so you can see two work areas at once. Use the ⧉ button on the tab name, double-click the tab, or right-click it and choose "Open in separate window". Most panels have their own ⧉ at the top right.



Mini Tools open in a floating window over the Arrange tab.

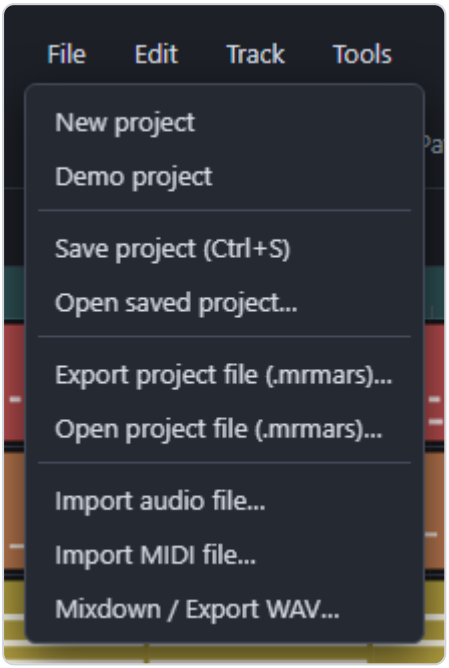
A floating window can be maximized, docked or closed from its title bar. When it closes, its content returns to its place. Window positions are remembered.

## Language

The default language is English. On first start the app uses your browser's language if it is available, and English if it is not. To choose by hand, press the 🌐 button beside the menus. "Automatic" returns to the browser language.

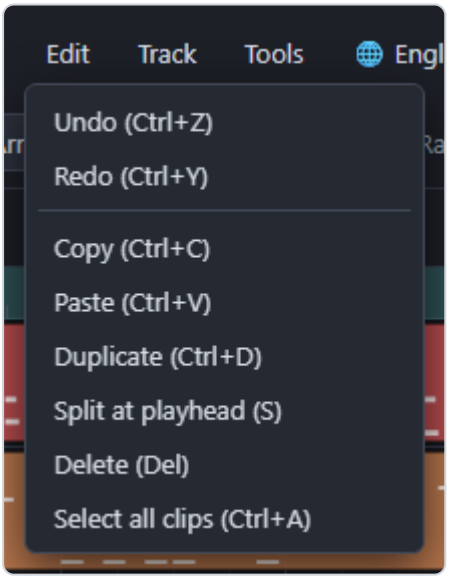
# 3. Menus

## File



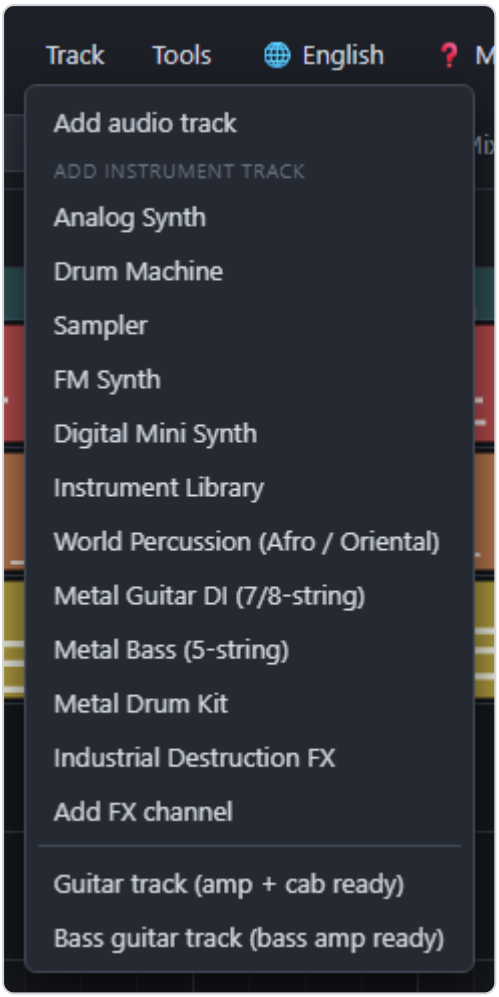

- **New project:** starts an empty project.
- **Demo project:** loads the example project.
- **Save project (Ctrl+S):** saves under a name inside the app.
- **Open saved project…:** lists the projects saved inside the app.
- **Export project file (.mrmars)…:** makes one file containing the project and its audio, for backup or sharing.
- **Open project file (.mrmars)…:** opens such a file. Older `.orbit` files also open.
- **Import audio file… / Import MIDI file…:** adds a file to the project.
- **Mixdown / Export WAV…:** see chapter 16.

## Edit



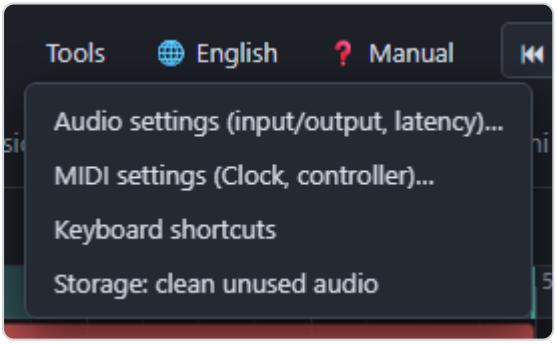

- **Undo, Redo:** `Ctrl` + `Z`, `Ctrl` + `Y`.
- **Copy, Paste, Duplicate:** `Ctrl` + `C`, `V`, `D`. These act on the selected clips.
- **Split at playhead:** `S` cuts the selected clip at the playhead.
- **Delete:** `Del` removes the selected clips.
- **Select all clips:** `Ctrl` + `A`.

## Track



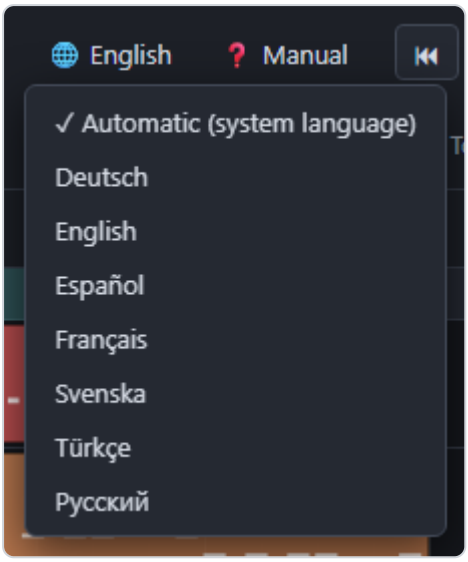

- **Add audio track:** for recording or importing audio.
- **Add instrument track:** choose an instrument from the list. These tracks hold MIDI.
- **Add FX channel:** a shared effect return that other tracks can send to.
- **Guitar track:** an audio track with an amp and cab ready.
- **Bass guitar track:** an audio track with a bass amp ready.

## Tools




- **Audio settings:** see chapter 16.
- **MIDI settings:** MIDI Clock and connected ports.
- **Keyboard shortcuts:** shows the list of shortcuts.
- **Storage: clean unused audio:** deletes recordings that no project uses. It cannot be undone.
- **Add to home screen:** appears where the browser supports installing the app.

## 🌐 Language




Lists Automatic and the seven languages. A tick marks the current choice. The app reloads when you pick a language, so save your work first.

## ❓ Manual

Opens this manual in a new window. It is available in English, whatever language the app is using. The About window has the same link. When the app is installed, the manual also opens offline.

## About




The About window shows the version, audio engine details, loaded plugin count, language and data folder.

# 4. Arrange tab

This is the main workspace, where recordings and MIDI clips are placed along a timeline.

## Toolbar

To add tracks or import audio, use the **Track** and **File** menus. To split a clip, use **Edit** or press `S`.

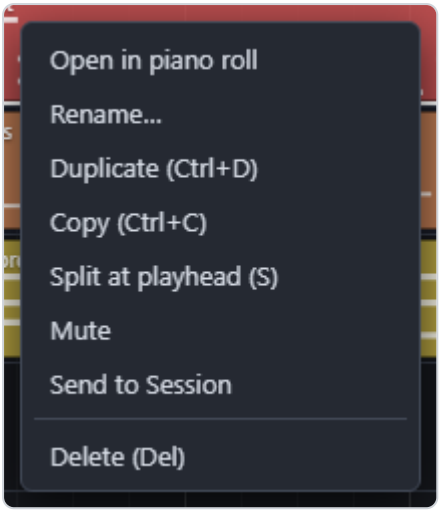
| Control | Function |
|---|---|
| Grid | Snap value: Off, 1 Bar, 1/4, 1/8, 1/16 or 1/32. |
| Zoom | Horizontal zoom. `Ctrl` + mouse wheel does the same. |

## Track header buttons

| Button | Function |
|---|---|
| M | Mute the track. |
| S | Solo the track. |
| ● | Arm the track for recording. |
| e | Open the instrument panel (instrument tracks). |
| 🎧 | Monitor the input for live playing (audio tracks). |
| A | Show or hide automation lanes. |
| Slider | Track volume. |

## Working with clips

- **Move:** drag a clip. `Alt` + drag copies it. `Shift` + drag ignores the grid.
- **Open:** double-click a MIDI clip to open the piano roll.
- **Loop region:** drag in the ruler.
- **Recording again** over a clip keeps the old one as a muted take, so you can compare takes.



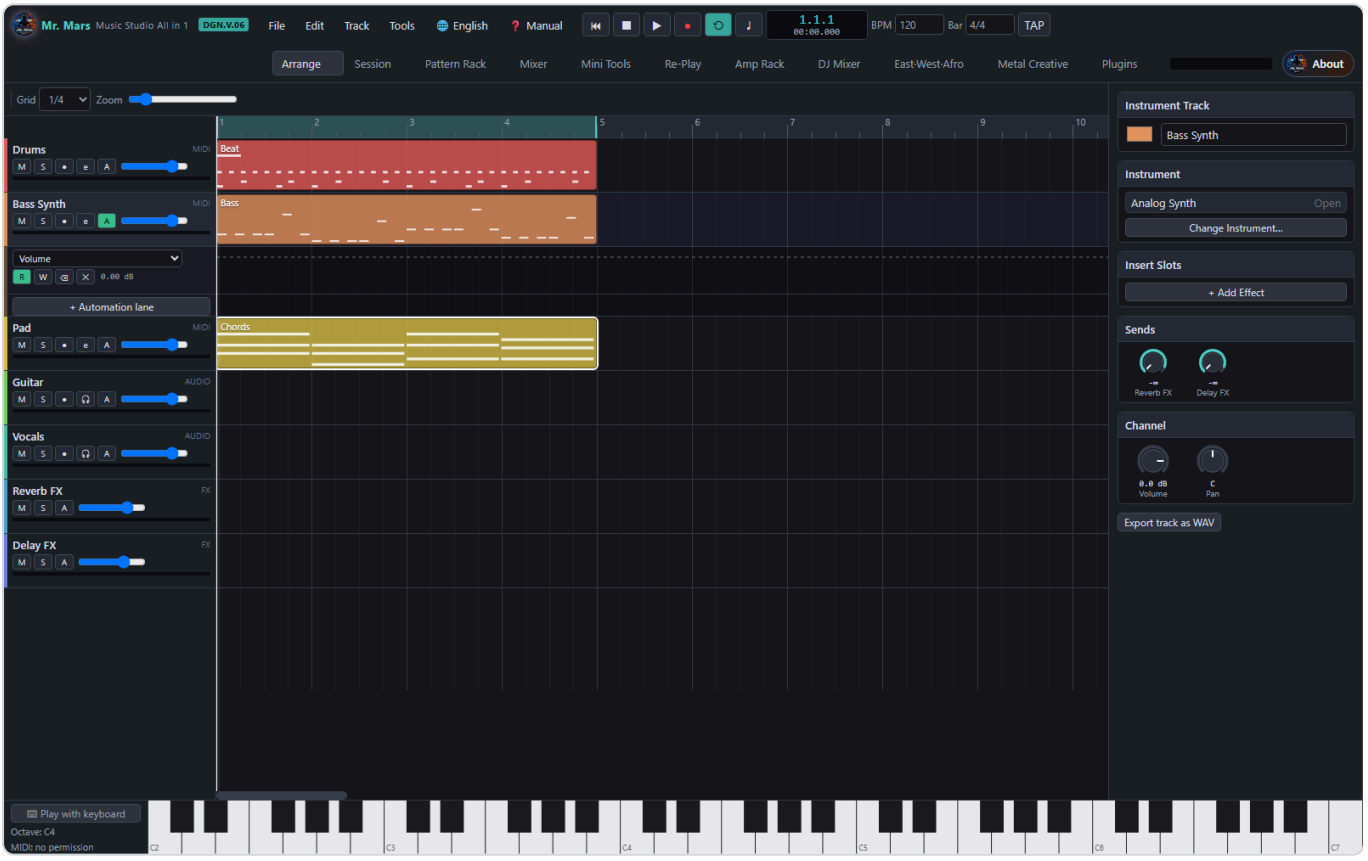

Right-click menu of a MIDI clip.

Right-click a clip for its menu. MIDI clips offer the items shown here. Audio clips add audio tools:

- **Split into stems:** vocals, drums, bass and melody. The result is approximate.
- **Fit to tempo:** changes the length and keeps the pitch.
- **Pitch shift:** changes the pitch and keeps the length.
- Reverse, normalize and fades.

**Send to Session** places the clip in the Session grid.

## Automation lanes



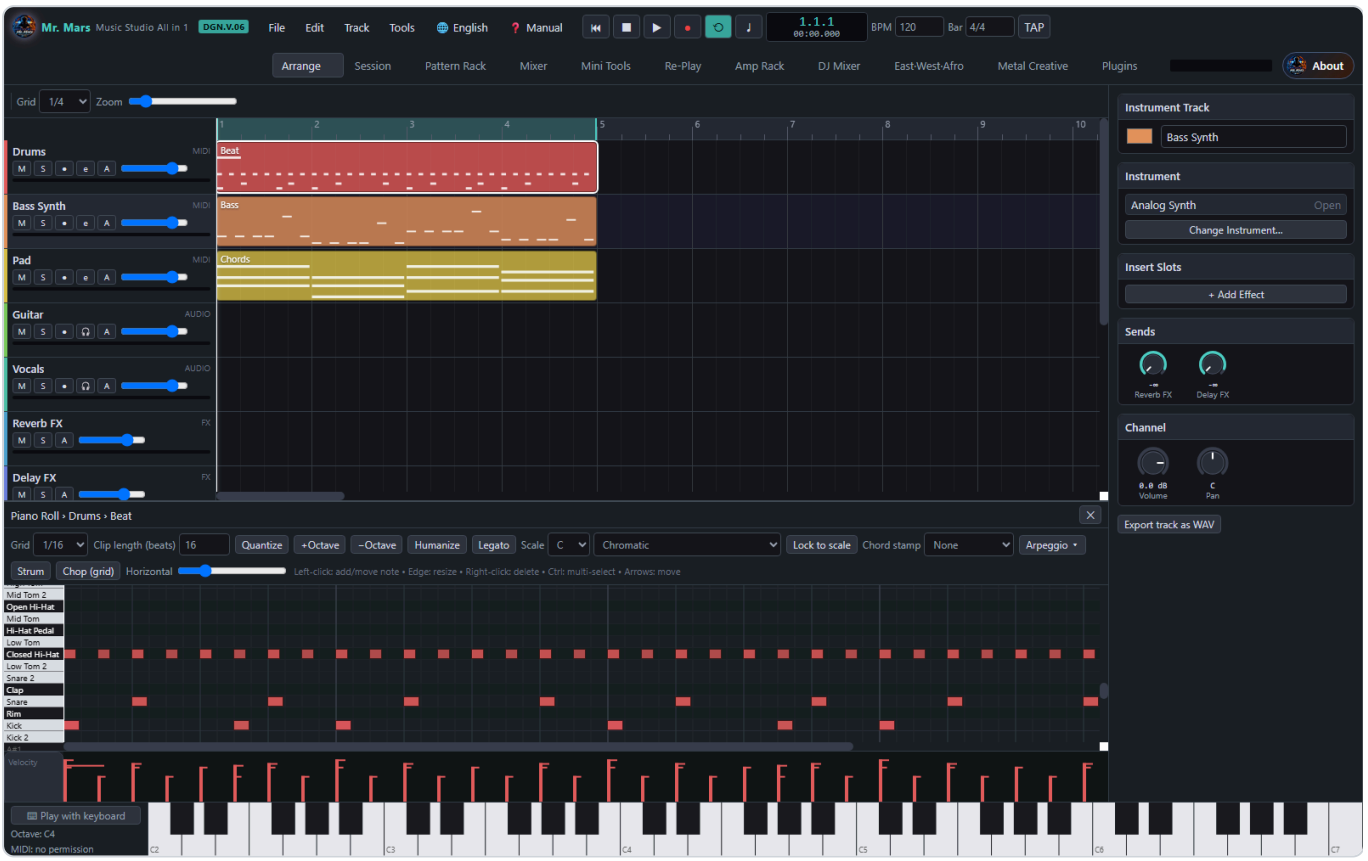

A Volume automation lane open under the Bass Synth track, with the track's details in the Inspector.

Press **A** on a track to open its lanes. Each lane can control Volume, Pan or any parameter of a device on the track. Use "+ Automation lane" to add more.

- **Add a point:** click empty space.
- **Move a point:** drag it.
- **Delete a point:** double-click or right-click it.
- **R (Read):** plays the curve.
- **W (Write):** records your control moves during playback.

Automation also applies to mixdown and stem export. The master channel and Amp Rack have no automation.

## Piano roll




The piano roll opens under the timeline. This clip is on a drum track, so rows show drum names.

**Left-click** adds or moves a note, **dragging the edge** resizes it, and **right-click** deletes it. `Ctrl` selects several notes and the arrow keys move them. The lane at the bottom sets velocity.

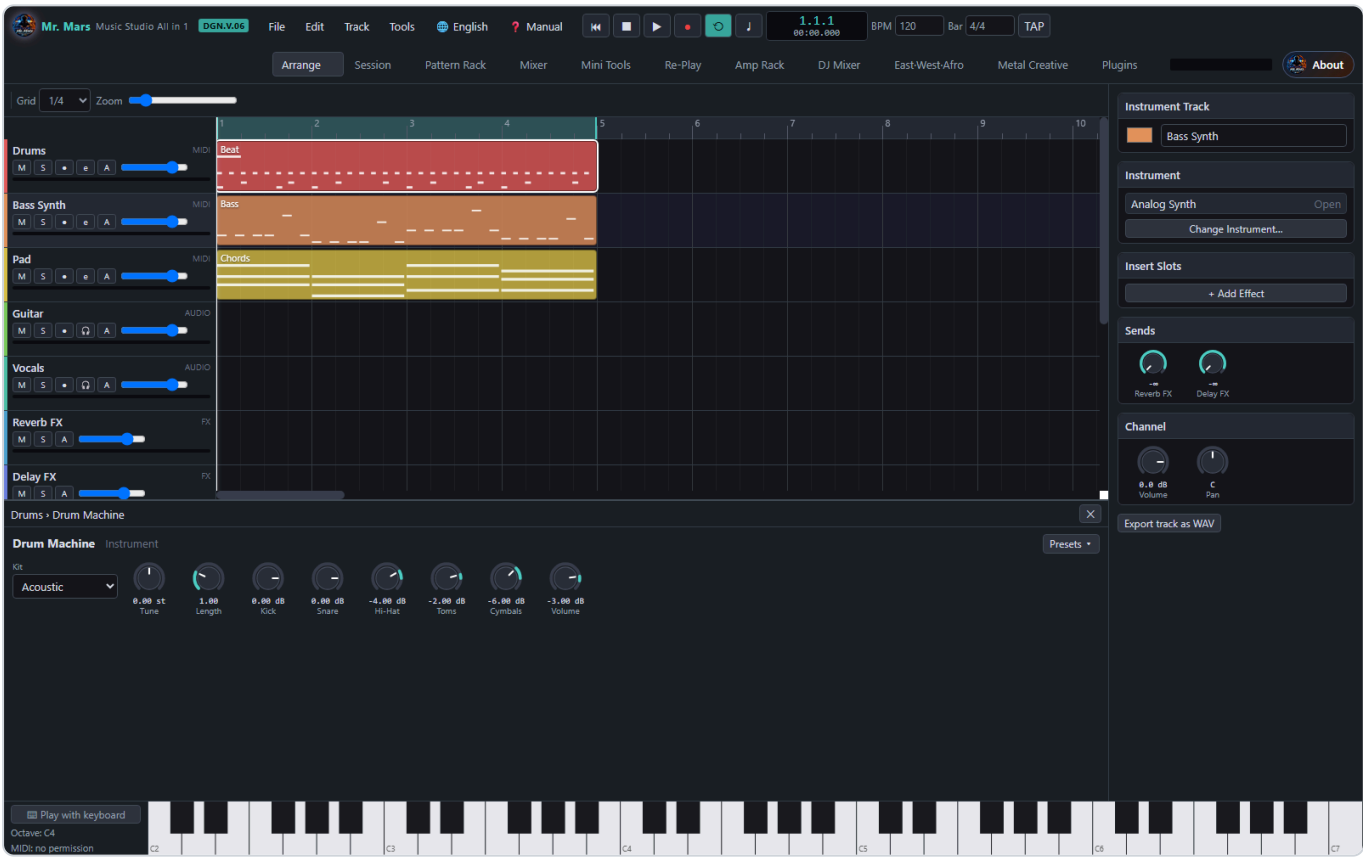
| Tool | Function |
|---|---|
| Grid, Clip length | Note snap value, and clip length in beats. |
| Quantize ( `Q` ) | Moves notes onto the grid. |
| +Octave, −Octave | Moves notes up or down an octave. |
| Humanize | Adds small timing and velocity changes. |
| Legato | Extends each note to the next one. |
| Scale, Lock to scale | Highlights a scale (major, minor, modes, pentatonic, blues, maqams) and keeps new notes on it. |
| Chord stamp | Adds a whole chord with one click. |
| Arpeggio, Strum | Turns a selected chord into an arpeggio or a strum. |

| Tool | Function |
| --- | --- |
| Chop (grid) | Cuts notes into grid-sized pieces. |

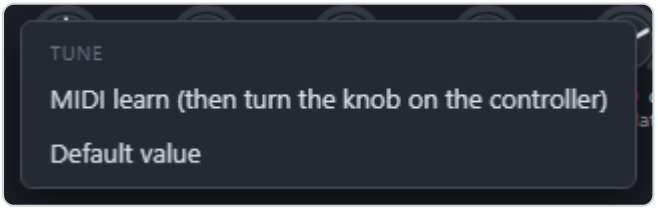
## Inspector and devices

Select a track to see its details on the right:

- **Name and colour.**
- **Instrument:** "Open" shows its panel and "Change Instrument..." replaces it.
- **Insert Slots:** effects in order. Use "+ Add Effect" to add one.
- **Sends:** how much signal goes to each FX channel.
- **Channel:** volume and pan.
- **Export track as WAV.**



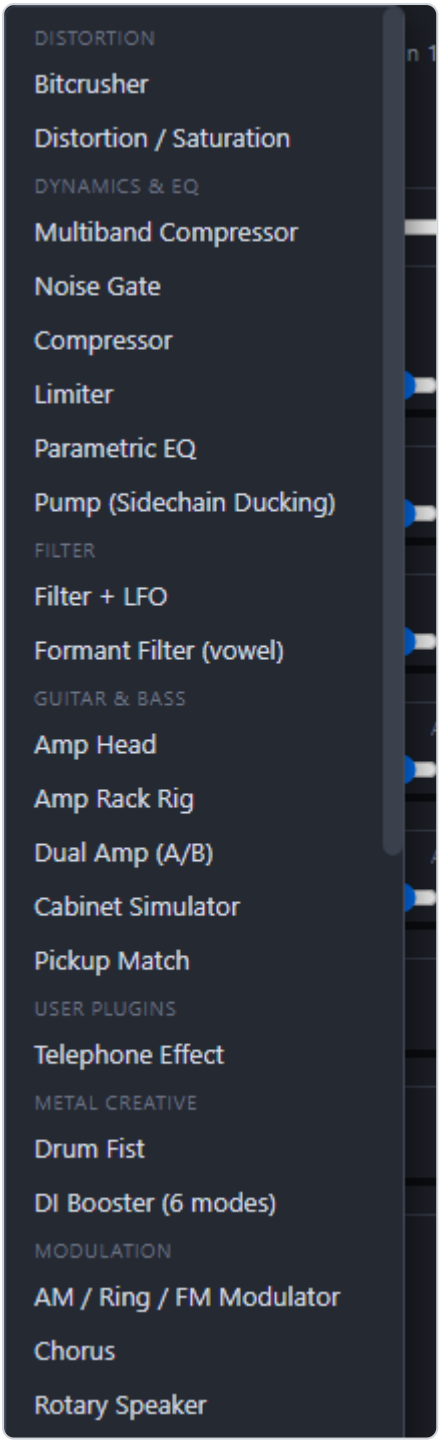

A device panel opens under the timeline. "Presets" loads and saves its settings.




Right-click a knob.

Drag a knob to change it. Right-click it for two options:

- **MIDI learn:** turn a knob on your controller to link it.
- **Default value:** resets the knob.




Part of the "+ Add Effect" list.

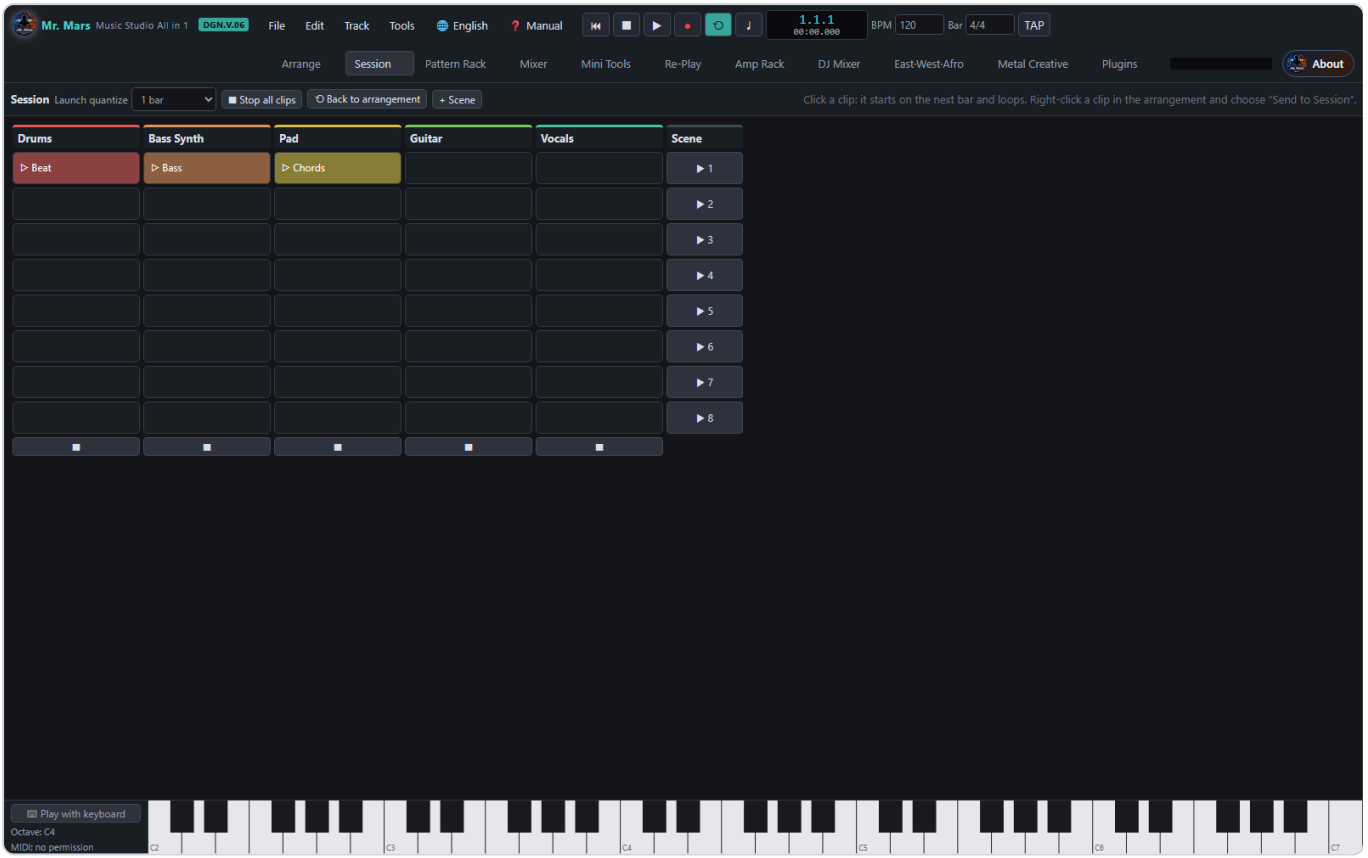
## Effects

- **Distortion:** Bitcrusher, Distortion / Saturation.
- **Dynamics & EQ:** Multiband Compressor, Noise Gate, Compressor, Limiter, Parametric EQ, Pump.
- **Filter:** Filter + LFO, Formant Filter.
- **Guitar & Bass:** Amp Head, Amp Rack Rig, Dual Amp, Cabinet Simulator, Pickup Match.
- **Metal Creative:** Drum Fist, DI Booster.
- **Modulation:** AM / Ring / FM, Chorus, Rotary Speaker, Flanger, Auto Pan, Phaser, Tremolo.
- **Pedals:** Clean Boost, Distortion, Fuzz, Graphic EQ, Compressor, Sub Octave, Overdrive, Wah.
- **Pitch and Performance:** Pitch Shifter, XY Effect Pad.
- **Studio:** FET and Opto Compressors, Cassette / Tape Machine, Saturator.
- **Creative:** Envelope Filter, Vinyl (Lo-Fi), Exciter, De-esser, Harmonizer, Trance Gate, Vocoder.
- **Space:** Reverb, Shimmer Reverb, Stereo Width.
- **Utility and Time:** Gain, Spectrum Analyzer, Delay.

## Instruments

Electric Guitars, Electric Basses, Fretless Electric Guitar (Turkish), Turkish Instruments, Bowed Strings, Viking Instruments, Analog Synth, Drum Machine, Recorded Drum Kit, Sampler, FM Synth, Digital Mini Synth, Instrument Library, World Percussion, Metal Guitar DI, Metal Bass, Metal Drum Kit and Industrial Destruction FX.

# 5. Session tab

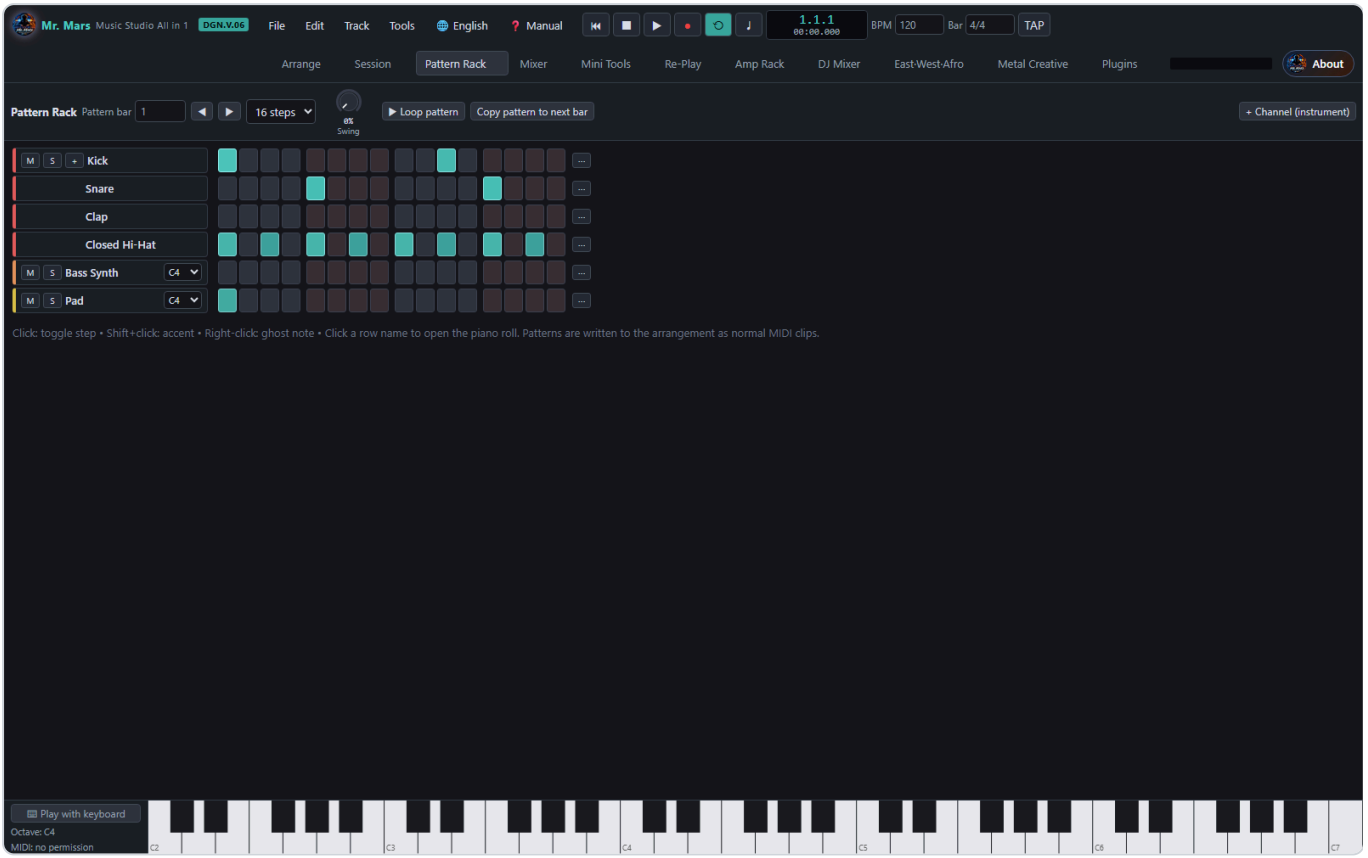



Three clips sent from the arrangement are ready in scene 1.

Session is a grid for launching clips live. Columns are tracks and rows are scenes.

- **Get clips here:** right-click a clip in Arrange and choose "Send to Session".
- **Launch a clip:** click it. It starts at the next launch point and loops.
- **Launch a scene:** press ▶ in the Scene column to start that whole row.
- **Stop:** ■ under a column stops that track. "■ Stop all clips" stops everything.
- **Launch quantize:** 1 bar, 1/2 bar, 1 beat or Immediately.
- **Back to arrangement:** returns playback to the Arrange timeline.
- **+ Scene:** adds a row.

# 6. Pattern Rack tab

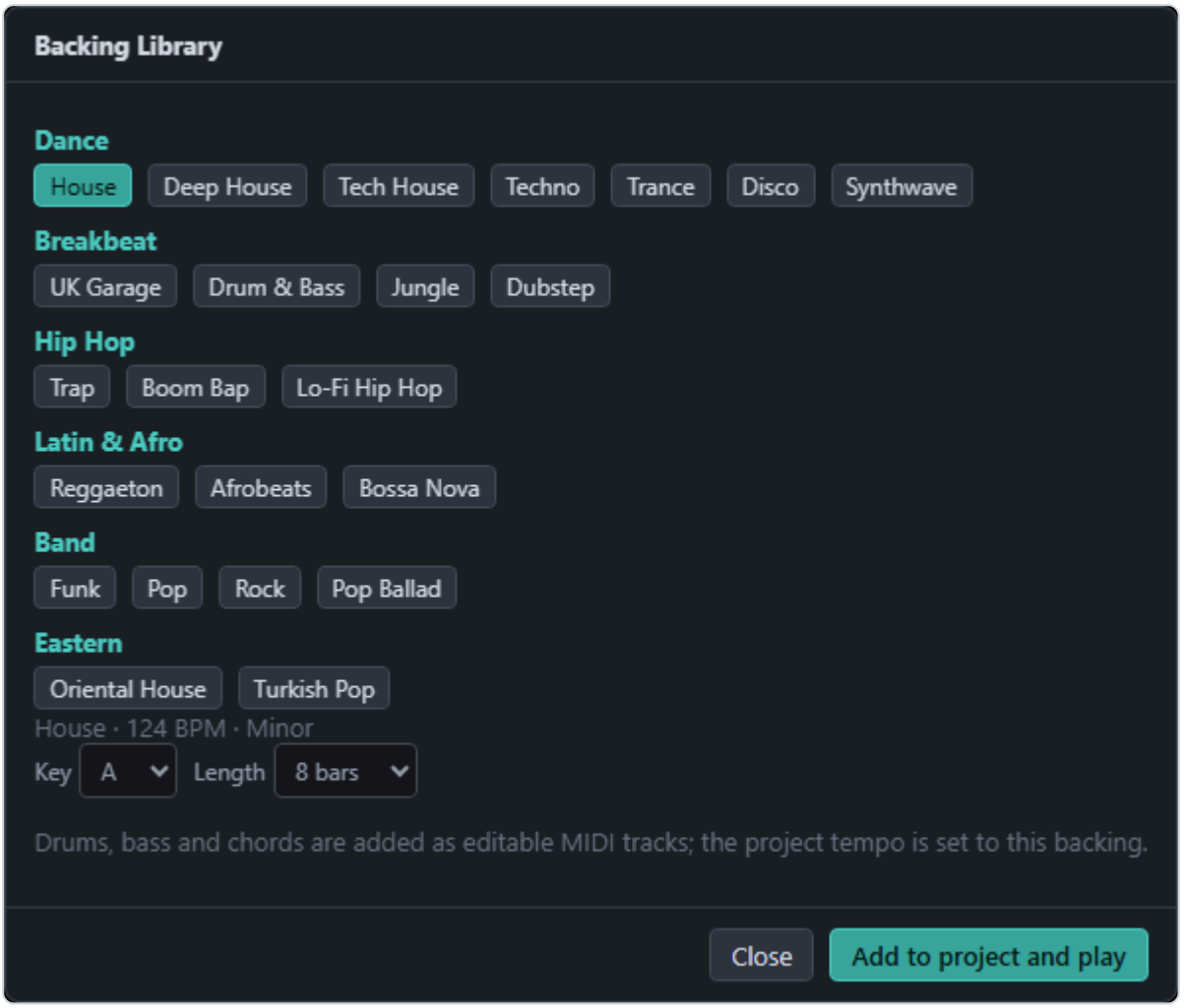



Step patterns for drum rows and instrument tracks.

A step sequencer. Each lit square plays a note on that step, and patterns are written to the arrangement as normal MIDI clips.

- **Click:** turn a step on or off.
- **Shift + click:** accent. **Right-click:** ghost note.
- **Click a row name:** open it in the piano roll.
- **Pattern bar, ◀ ▶:** the bar you are editing.
- **Steps:** 16, 32 or 64. **Swing:** shuffles the timing.
- **▶ Loop pattern:** repeats the current bar.
- **Copy pattern to next bar:** builds longer parts quickly.
- **+ Channel (instrument):** adds a new instrument row.
- **+ on a drum row:** shows more drum sounds; press it again for every sound of that device, such as the Eastern percussion. **⋯ at the row end:** fill options.
- **Note menu on instrument rows:** chooses the pitch of the row.

# Backing library




Backings button: pick a genre, a key and a length.

The **Backings** button at the top right of the Pattern Rack opens 23 ready genres in six groups. Choose a genre, a key and 4, 8 or 16 bars, then press **Add to project and play**.

Drums, bass and chords are added after your last clip as normal MIDI tracks, so you can edit every note, change the instruments or delete a part. The project tempo changes to the tempo of the backing, and a loop is set around it. Edit ▸ Undo removes the backing and restores the previous tempo and loop.

# 7. Mixer tab



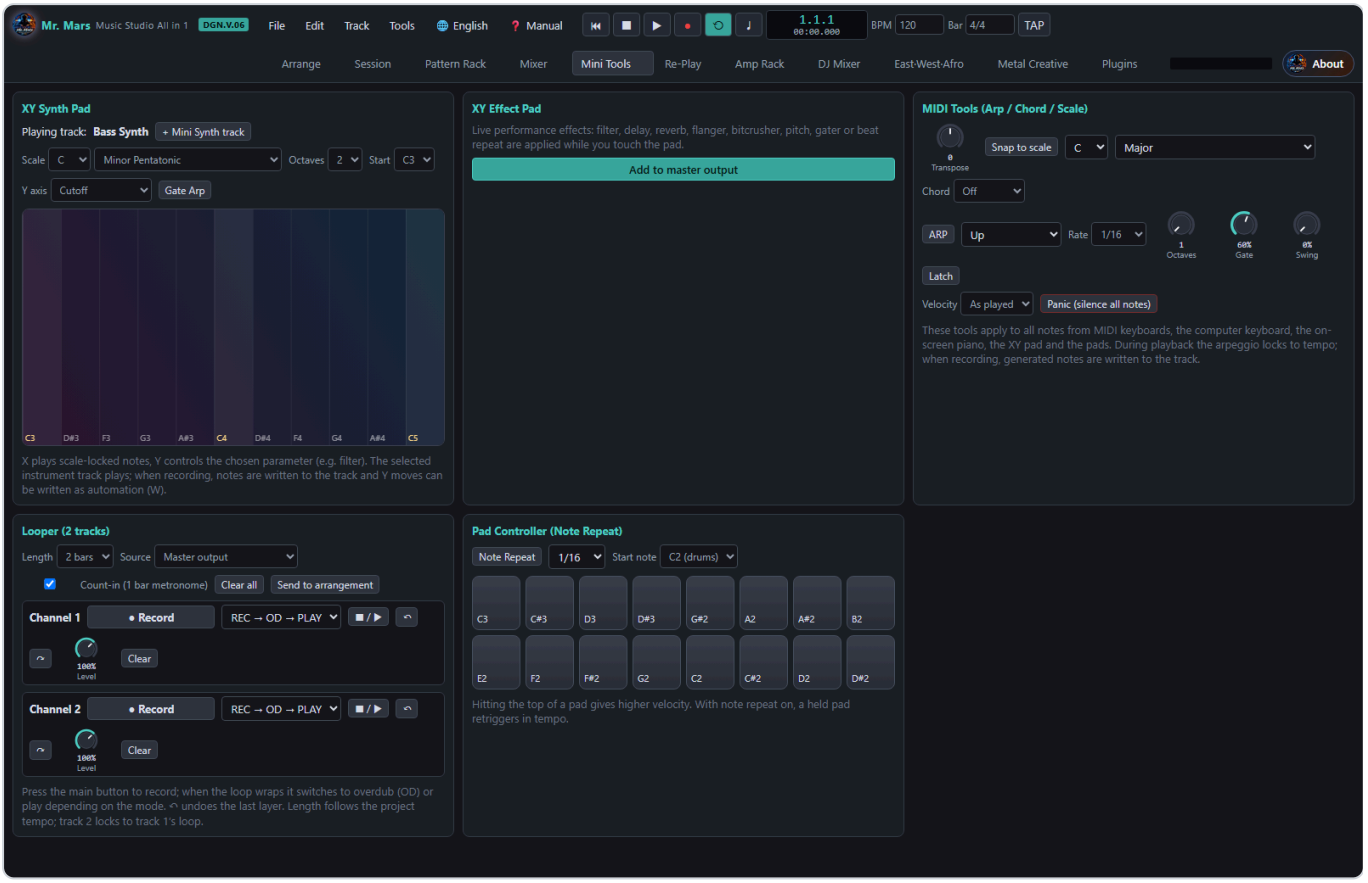

One channel strip per track, with the Master on the right.

From top to bottom, each channel strip has:

- **Name** in the track colour.
- **Instrument** (marked ♪) and **insert effects**. Click one to open it. **+** adds an effect.
- **Send knobs** for each FX channel.
- **Pan** knob.
- **Fader** and **level meter**.
- **M, S, ●:** mute, solo and record arm.

The demo project includes Reverb FX and Delay FX channels. Turn up a send on any track to feed them. The **MASTER** strip holds the final effects, such as the Limiter.

# 8. Mini Tools tab




Five small performance tools.

## XY Synth Pad

Horizontal position plays notes locked to the chosen scale. Vertical position controls the parameter chosen under "Y axis". It plays the selected instrument track, and "+ Mini Synth track" makes a new one. While recording, notes are written to the track, and Y moves can be written as automation with W. **Gate Arp** repeats the note in tempo while held.

## XY Effect Pad

A live effect applied while you touch the pad: filter, delay, reverb, flanger, phaser, bitcrusher, pitch, gater or beat repeat. "Add to master output" places it on the master.

## MIDI Tools

These act on all notes from MIDI keyboards, the computer keyboard, the on-screen piano, the XY pad and the pads.

- **Transpose** and **Snap to scale**.
- **Chord:** one key plays a whole chord.
- **ARP:** arpeggiator with pattern, rate, octaves, gate and swing. **Latch** keeps it running after you release the keys.

- **Velocity:** As played, Fixed, Soft or Hard.
- **Panic:** silences all notes.

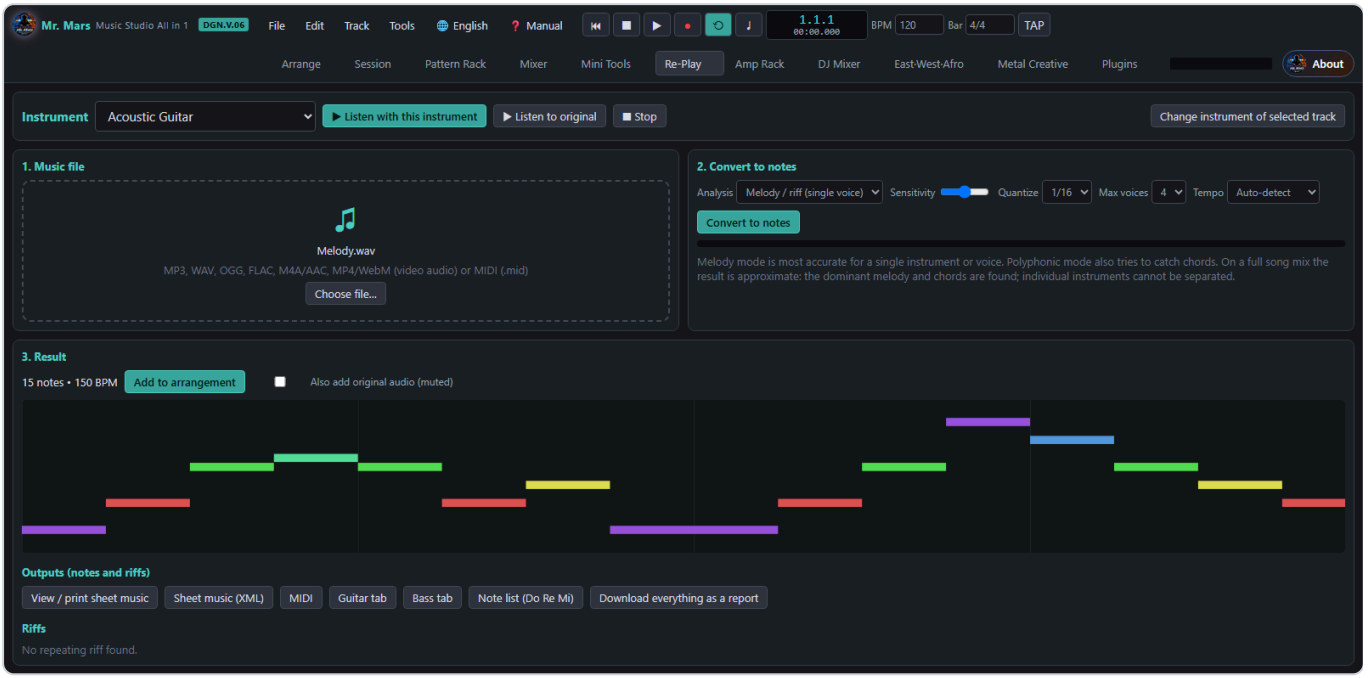
## Looper (2 tracks)

Choose the length and source (master output, or audio input for mic or guitar). Press **Record**. When the loop wraps it moves to overdub or play, depending on the mode. ↶ and ↷ undo and redo layers. "Send to arrangement" turns the loops into clips. Length follows the project tempo, and track 2 locks to the loop of track 1.

## Pad Controller

Sixteen pads that play the selected instrument. Hitting the top of a pad gives higher velocity. With **Note Repeat** on, a held pad retriggers in tempo at the chosen rate.

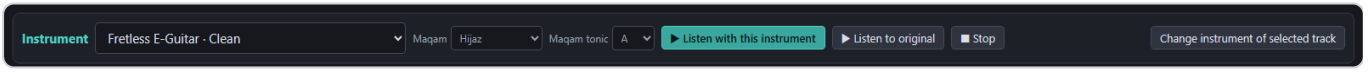
# 9. Re-Play tab




A short melody converted to notes, with the result and its outputs.

Re-Play turns a music file into notes and plays them with another instrument.

## Electric and fretless guitars




A fretless tone with its Maqam and Maqam tonic lists.

**Electric Guitar** has 25 tones: single-coil neck, in-between and bridge, blues, surf, rockabilly, 80s clean, jazz hollow body, semi-hollow blues, humbucker clean, classic rock, hard rock, solo lead, modern metal rhythm, 60s fuzz and electro-acoustic, plus nine European metal styles: Swedish death, Gothenburg melodic death, Norwegian black metal, British heavy metal, German thrash, German power metal solo, Neue Deutsche Härte, British doom and Viking / pagan metal. On tones with palm mute, notes with velocity under 70 are muted.

**Electric Bass** has 9 tones: finger, pick, classic heavy metal, thrash, death metal, black metal, doom, pagan / folk metal and modern five-string metal. The distorted tones keep a clean low end under the distortion.

**Bowed Strings** adds electric violin (clean and metal), folk metal violin, distorted metal cello, symphonic metal strings, hurdy-gurdy (acoustic and electric metal, with drone strings and wheel buzz), nyckelharpa, Hardanger fiddle, gusle and crwth. Instruments with drones play them with every note, so they suit a fixed key: C and G on the hurdy-gurdy.

**Fretless Electric Guitar (Turkish)** has 7 tones: clean, oud tone, saz tone, soft jazz, Anatolian rock, distortion solo and ney tone. They slide between adjacent notes and have a wide, delayed vibrato. Notes played together do not slide.

- **Maqam and Maqam tonic:** appear for Turkish instruments. Choose one to play that track in 53-comma pitches; other tracks do not change. Keys with no match in the maqam stay at their normal pitch.
- **Convert to notes** still finds whole semitones. The maqam then moves those notes to its comma pitches, so it does not capture the exact pitches of the original recording.

**Viking Instruments** has 11 instruments: Tagelharpa, Viking lyre, Kravik lyre, bukkehorn, lur, war horn, bone flute, Viking pan flute, willow flute, jaw harp and Viking frame drum. No recordings of Viking-age music survive, so these are approximate models of instrument types known from finds and tradition. The tagelharpa keeps two drone strings on A and E.

Each instrument appears in one list. In the instrument menu, the Instrument Library holds only the groups that have no device of their own. Hand percussion is also in one place: darbuka, bendir, def and erbane in the Drum Machine; djembe, conga, bongo, shaker and clave in World Percussion. Old projects keep playing any sound removed from a list. Core kit pieces such as kick, snare and hi-hat stay in each drum kit, because each kit is its own sound set.

The same tones are in the instrument menu as **Electric Guitars**, **Electric Basses**, **Bowed Strings**, **Viking Instruments** and **Fretless Electric Guitar (Turkish)**. The **Drum Machine** also has Turkish and Middle Eastern percussion on notes 65 to 86: darbuka, bendir, def, erbane, riq, kudüm, davul, nakkare, finger cymbals, spoons, çarpara, tombak, daf and mazhar. Set their level with the **Eastern Percussion** knob. **Turkish Instruments** is there too, with bağlama, cura, oud, kanun, cümbüş, ney, Turkish clarinet, zurna, kemençe, duduk, yaylı tambur, tanbur, divan sazı, kaval, mey, sipsi, tulum, kabak kemane, santur and electric bağlama. The fretless and Turkish Instruments devices start in Hijaz on A. Its **Glide** knob sets the slide time, and **Expression** sets the vibrato width.

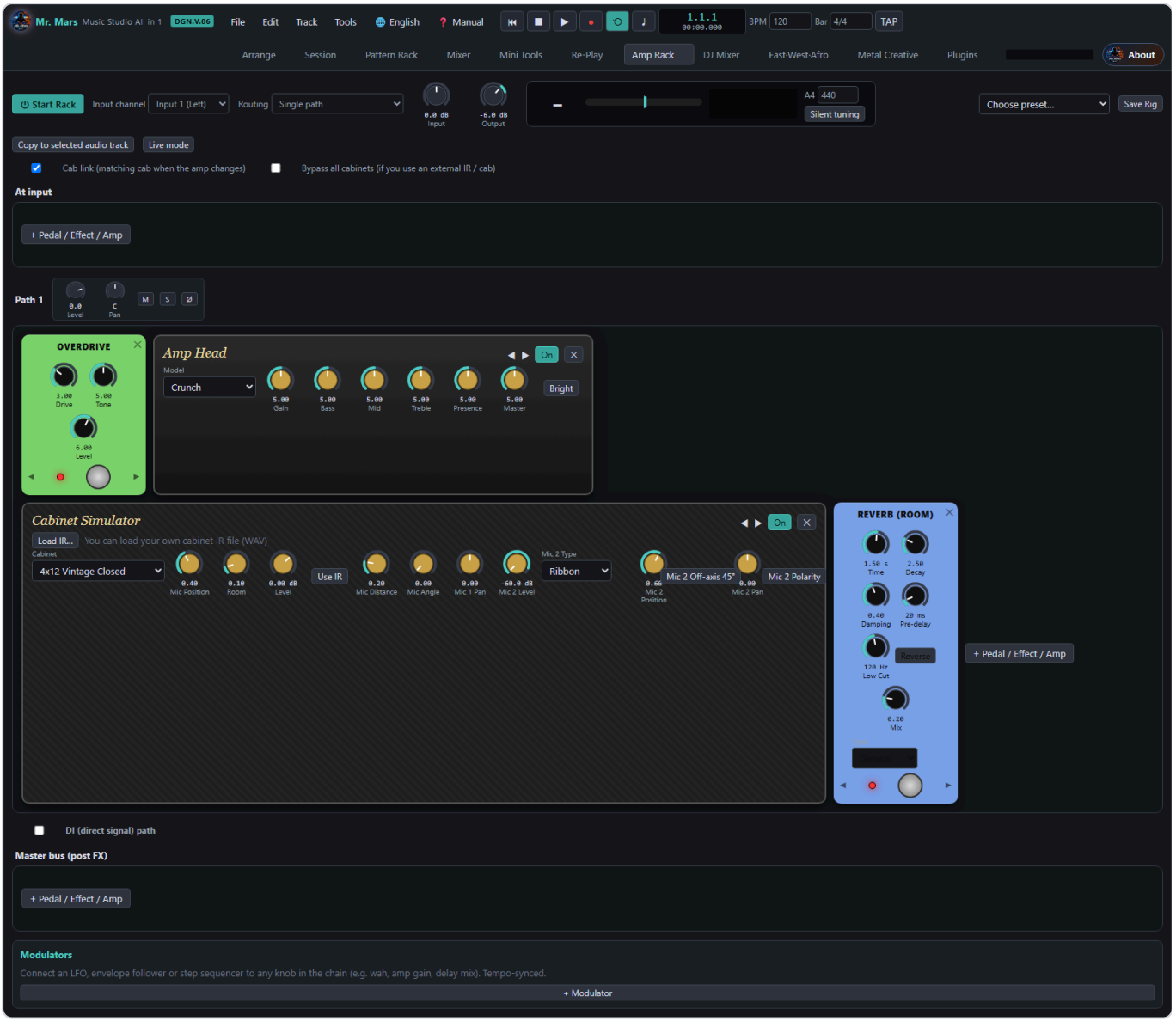
## Using Re-Play

1. **Music file:** drag in or choose an MP3, WAV, OGG, FLAC, M4A/AAC, MP4/WebM or MIDI file.
2. **Convert to notes:** set the options and press the button.
    - **Analysis:** Melody / riff for a single voice, or Polyphonic for chords.
    - **Sensitivity, Quantize, Max voices.**
    - **Tempo:** detect it or use the project tempo.
3. **Result:** check the notes, then listen or add them to the project.

- **Instrument:** choose from 125 instruments: guitars, Turkish and world instruments, keys, winds, strings and synths.
- ▶ **Listen with this instrument** and ▶ **Listen to original** let you compare.
- **Add to arrangement:** creates a track. It can also add the original audio, muted.
- **Outputs:** view or print sheet music, Sheet music (XML), MIDI, Guitar tab, Bass tab, Note list (Do Re Mi), or everything as a report.
- **Riffs:** repeating riffs are listed when found.

> Melody mode is the most accurate, for one instrument or voice. On a full song the result is approximate: the main melody and chords are found, but separate instruments cannot be pulled apart.

# 10. Amp Rack tab




A guitar chain: overdrive pedal, amp head, cabinet and reverb.

## Getting sound

1. Connect your instrument. See chapter 15.
2. Press ⏻ **Start Rack** and allow the microphone.
3. Choose the **Input channel**: Input 1, Input 2 or Stereo.
4. Pick a preset from "Choose preset...".

## Top controls

| Control | Function |
|---|---|
| Routing | Single path, 2-way splitter, 3-way splitter (wet/dry/wet) or Parallel stereo (2 inputs). |
| Input, Output | Input and output level of the rack. |
| Tuner, A4, Silent tuning | Tuner display, reference pitch, and muting the output while you tune. |
| Save Rig | Saves the whole chain as a preset. |
| Copy to selected audio track | Puts the chain on a track, so you can record through it. |
| Live mode | A preset view for playing live. MIDI Program Change can switch presets. |
| Cab link | Selects a matching cab when the amp changes. |
| Bypass all cabinets | For when you use an external IR or cabinet. |

## The signal chain

- **At input:** devices before the split, for example a gate or compressor.
- **Path:** one row per path, with its own Level, Pan, M, S and Ø (invert phase).
- **DI (direct signal) path:** adds the dry signal in parallel.
- **Master bus (post FX):** devices after all paths.
- **+ Pedal / Effect / Amp:** adds a device. The ◀ ▶ arrows, or dragging, change the order.

## Amp and cab

**Amp Head:** choose a model and set Gain, Bass, Mid, Treble, Presence and Master. Amps are named by style:

- **Clean:** Clean, Bright Clean, Clean Reverb, Clean Jazz.
- **Crunch:** Crunch, Classic Crunch, Bright Crunch, Vintage Combo, Warm Combo, Soft Overdrive.
- **Lead:** Lead, Smooth Lead, Hard Rock.
- **Metal:** High Gain, Modern Hi-Gain, Heavy Metal, Tight Metal, Modern Metal.
- **Bass:** Bass Tube, Bass Modern, Bass Classic, Bass Hi-Fi, Bass Overdrive.

**Cabinet Simulator:** choose a cabinet and place up to two microphones (position, distance, angle, pan, level, type and polarity). To use your own cabinet response, press **Load IR...** with a WAV file and turn on **Use IR**.

## Modulators

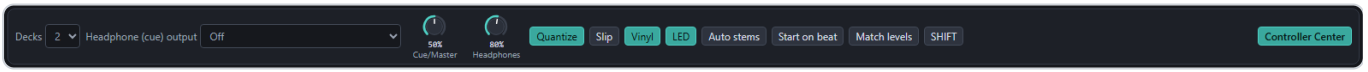
**+ Modulator** connects an LFO, envelope follower or step sequencer to any knob in the chain, such as wah, amp gain or delay mix. They can follow the tempo.

# 11. DJ Mixer tab



Two decks with the mixer column between them. Further panels are below; scroll down to reach them.

## Top bar



| Control | Function |
|---|---|
| Decks | 2 or 4 decks. |
| Headphone (cue) output | Off; Split (left = master, right = headphones); or a 4-channel interface using outputs 3-4. |
| Cue/Master, Headphones | Headphone blend and level. |
| Quantize | Cues, loops and beat jumps snap to beats. |
| Slip | When you leave a loop, the track continues as if the loop never happened. |
| Vinyl | Touching the jog lets the record follow your hand, so you can scratch. Your controller's jog touch must be mapped. |
| LED | Lights your controller's button LEDs. |

| Control | Function |
|---|---|
| Auto stems | Splits every loaded track into stems. This takes a few seconds per track. |
| SHIFT | Second pad function, for touch screens. |
| Start on beat | PLAY waits for the playing deck's next beat and starts from its own beat. |
| Match levels | Brings the loudness of tracks closer together, up to 12 dB either way. |
| Controller Center | See Controller Center. |

## Deck

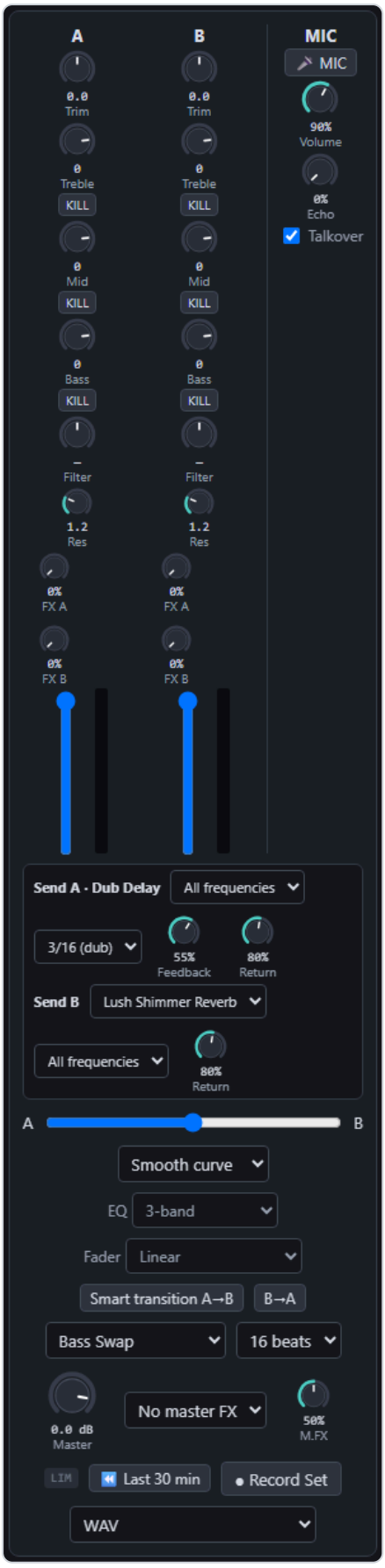



Deck A with a test track loaded.

| # | Control | Function |
|---|---|---|
| 1 | Title, X, Load... | Track name; crossfader side (A, Thru, B); load a file. |
| 2 | Overview | The whole track. Click to jump. |

| # | Control | Function |
|---|---|---|
| 3 | Waveform | Zoomed waveform in three colour bands, with beat markers. |
| 4 | Readout | BPM, key, time, time left and pitch. |
| 5 | Jog, ✋ | Nudge or move through the track. ✋ is jog touch for controllers, which scratches when Vinyl is on. |
| 6 | CUE, ▶, SYNC, KEY LOCK, 🎧 CUE | CUE: while stopped, sets the cue point; at the cue point, hold it to preview and let go to return; while playing, returns to the cue point and stops. Press ▶ during a preview to keep playing. ▶ plays; SYNC matches tempo and beats, including half or double tempo; KEY LOCK keeps pitch when the tempo changes; 🎧 CUE listens in headphones. |
| 7 | Split stems, Tone, Key Match | Split into vocals, drums, bass and melody; shift the key; match the other deck's key. |
| 8 | Jump, nudge, BPM | ◀◀ ▶▶ with a beat length (1 to 32); − + nudge while held; BPM… sets it by hand; ×2 / ÷2 corrects it. |
| 9 | Tempo, range | Tempo slider with a range of ±6, 8, 16 or 50%. |
| 10 | FX 1, 2, 3 | Three chained effects: Echo, Reverb, Flanger, Phaser, Bit Crush, Gater, Pitch, Chorus, Ring Mod or Shimmer. |
| 11 | Brake, Spinback, Reverse, TAP, Grid | Slow to a stop; spin back; play backwards; tap the BPM; shift the beat grid by 10 ms; "1." makes the current position beat 1. |
| 12 | Loop, pads | Loops from 1/4 to 32 beats, then the pad mode menu and pads. |

# Mixer column




- **Each channel:** Trim; Treble, Mid and Bass with KILL; Filter and its Res; FX A and FX B sends; channel fader and meter.
- **MIC:** switch, Volume, Echo and Talkover, which lowers the music while you talk.
- **Send A · Dub Delay:** frequency band, time, Feedback and Return.
- **Send B:** reverb type (Shimmer, Spring or Hall), frequency band and Return.
- **Crossfader:** blends A and B, with a Smooth or Sharp curve.
- **EQ:** 3-band or 4-band isolator.
- **Fader:** Linear, Rotary or Fast cut.
- **Smart transition A→B, B→A:** matches the tempo, starts on the bar and runs the chosen recipe over 8 to 64 beats. See Transition recipes.
- **LIM:** lights when the safety limiter is working. Turn the level down.
- ⏪ **Last 30 min:** downloads the recent mix, even if you never pressed record.
- **Master** level and **master effect** (M.FX).
- ● **Record Set:** records your mix as WAV or compressed.

## Transition recipes

Automix and Smart transition use the same recipes. Each one moves the crossfader, EQ, filter and effects in time with the beat.

| Recipe | What happens |
|---|---|
| Bass Swap | The incoming track fades in without bass. Halfway, the basses swap on the beat, so two basses never play together. |
| Fade | A plain crossfade. |
| Echo Out | Echo opens on the outgoing track, then the track drops out and its echo tails off while the new one comes in. |
| Filter Sweep | The outgoing track loses its lows while the incoming track opens from a low-pass filter. |
| Reverb Out | Like Echo Out, with reverb. |
| Tape Stop | The outgoing track slows to a stop, then the new one starts. |
| Cut | Switches at once. |

Echo Out and Reverb Out use the settings of Send A and Send B. Automix keeps the tempo unchanged when two tracks are more than 8% apart.

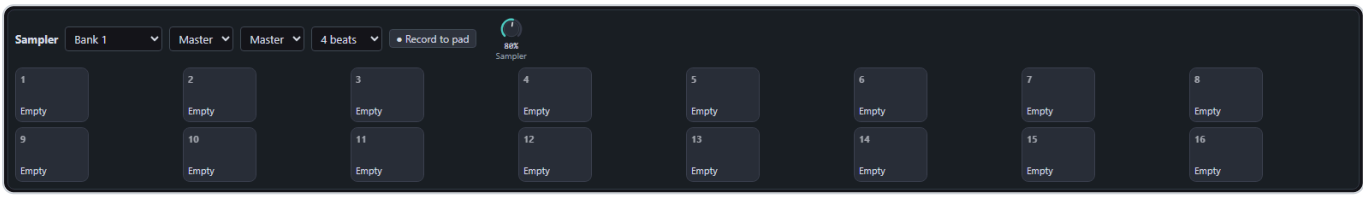
## Pad modes

Each deck has eight pads. The menu to their left chooses what they do. With a pad Shift function, hold `Shift` or use the SHIFT button.

| Mode | Function |
|---|---|
| Hot Cue, Hot Cue 9-16 | Click to set or jump. Shift + click deletes. |
| Loop Roll | A short loop while held. The track continues underneath. |
| Slicer | Splits the current bar into 8 slices. The held slice repeats. |
| Beat Jump | Jumps by a number of beats. |
| Loop+ | Loops by length, from 1/32 to 128 beats. |
| Cue Loop | Starts a loop from a cue point. |
| Key Cue | Plays from hot cue 1 in the chosen key. |
| Stems, Stem + FX | Turns stems on and off, or sends them to effects. |
| Manual Loop, Saved Loop | Loop in and out by hand; stored loops. |

| Mode | Function |
|---|---|
| Remix Points | Section starts found automatically. |
| Scratch | Eight ready scratch patterns. The track continues underneath. |

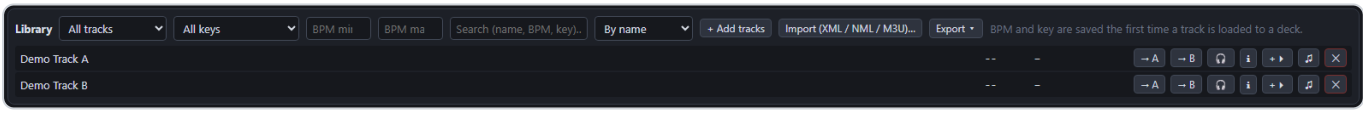
## Sampler




The bank list also has two ready banks: **Preset: DJ Effects** (riser, downlifter, impact, sub drop, air horn, siren, laser, zap, sweep, reverse cymbal, snare roll, vinyl stop and more) and **Preset: Recorded Drums** (real drum and percussion recordings in the public domain). Choosing one adds a new bank and fills its 16 pads; your own banks do not change.

Sixteen pads per bank, with as many banks as you need ("+ New bank…"). Load a sound onto a pad, and choose its trigger mode: On/Off, Stutter, Hold, Loop or Unmute. Output goes to the master or a deck. ● **Record to pad** records 4 to 32 beats from the master or a deck.

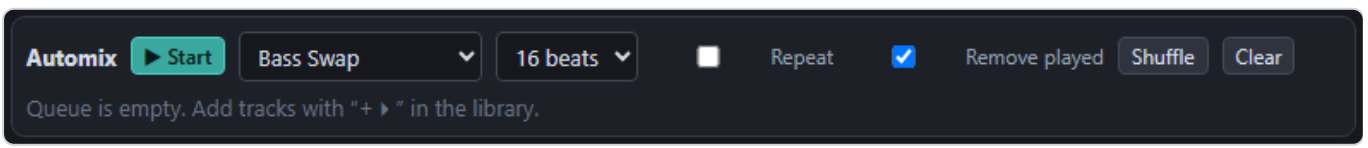
## Library




- **+ Add tracks:** adds files. Tags and covers are read automatically. BPM and key are saved the first time a track loads to a deck.
- **Filters:** playlist, compatible key, BPM range and text search.
- **Sort:** best match with the playing deck (key and BPM), name, artist, BPM, key, rating or date added.
- **Each track:** → A and → B load it; 🎧 prelistens; ℹ edits its information and rating; "+ ▸ " adds it to the Automix queue; ♫ adds it to a playlist; ✕ removes it.
- **Import:** DJ Library XML, DJ Collection NML or M3U, including BPM, key, beat grid, hot cues and playlists.
- **Export ▾:** DJ Library XML, or M3U8 for the whole library.

> Tracks are stored inside the app. Removing one from the library does not delete your original file.

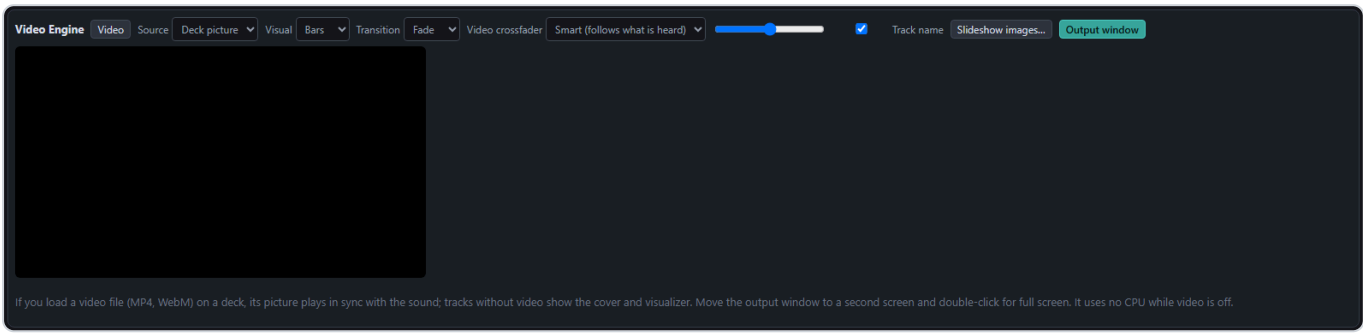
## Automix and History



**Automix** plays the queue by itself, using any transition recipe over 8 to 32 beats. "Remove played" takes finished tracks out of the queue, and Shuffle reorders it.

**History** lists the tracks you played. It can be downloaded as text or cleared.

## Video Engine

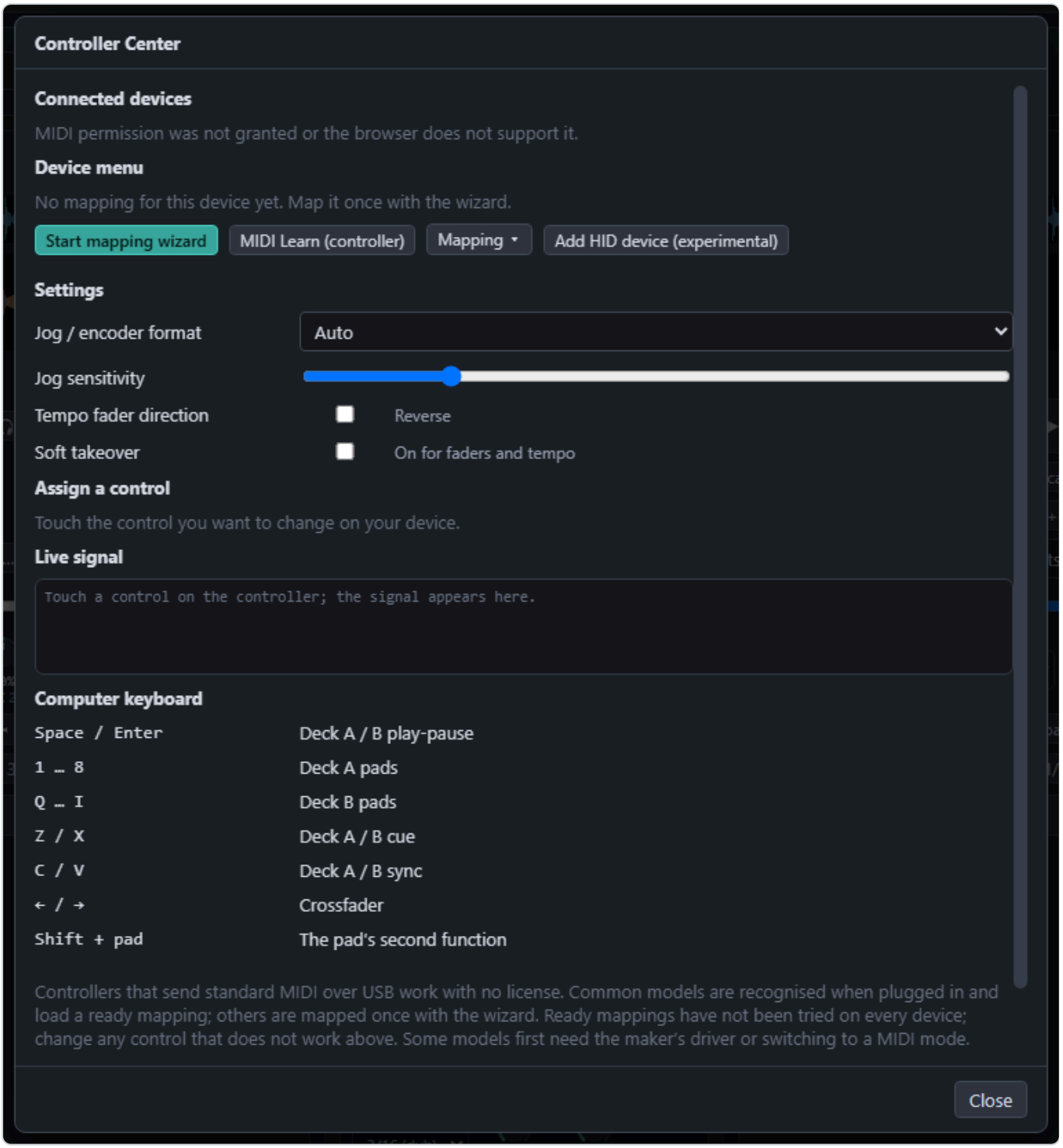



If a deck holds a video file (MP4, WebM), its picture plays in sync with the sound. Tracks without video show the cover and a visualizer.

- **Source:** Deck picture, Visualizer, Camera or Slideshow.
- **Visual:** Bars, Wave or Ring.
- **Transition:** Fade, Wipe or Zoom.
- **Video crossfader:** Smart, Linked to crossfader, or Manual.
- **Output window:** move it to a second screen and double-click for full screen.

It uses no CPU while Video is off.

# Controller Center




- **Connected devices:** what the app sees now.
- **MIDI Learn (controller):** click a control on screen, then move the control on your device. Press Finish when done.
- **Mapping ▾:** save mapping to file, load a mapping file, or delete all mappings.
- **Start mapping wizard:** asks you to move each control in turn and saves a profile per device.
- **Add HID device (experimental):** for models that use only their own protocol. Lights and high-resolution jog do not work on this path.
- **Back to ready mapping:** appears for recognised devices and undoes your changes.
- **Jog / encoder format:** Auto works for most devices.

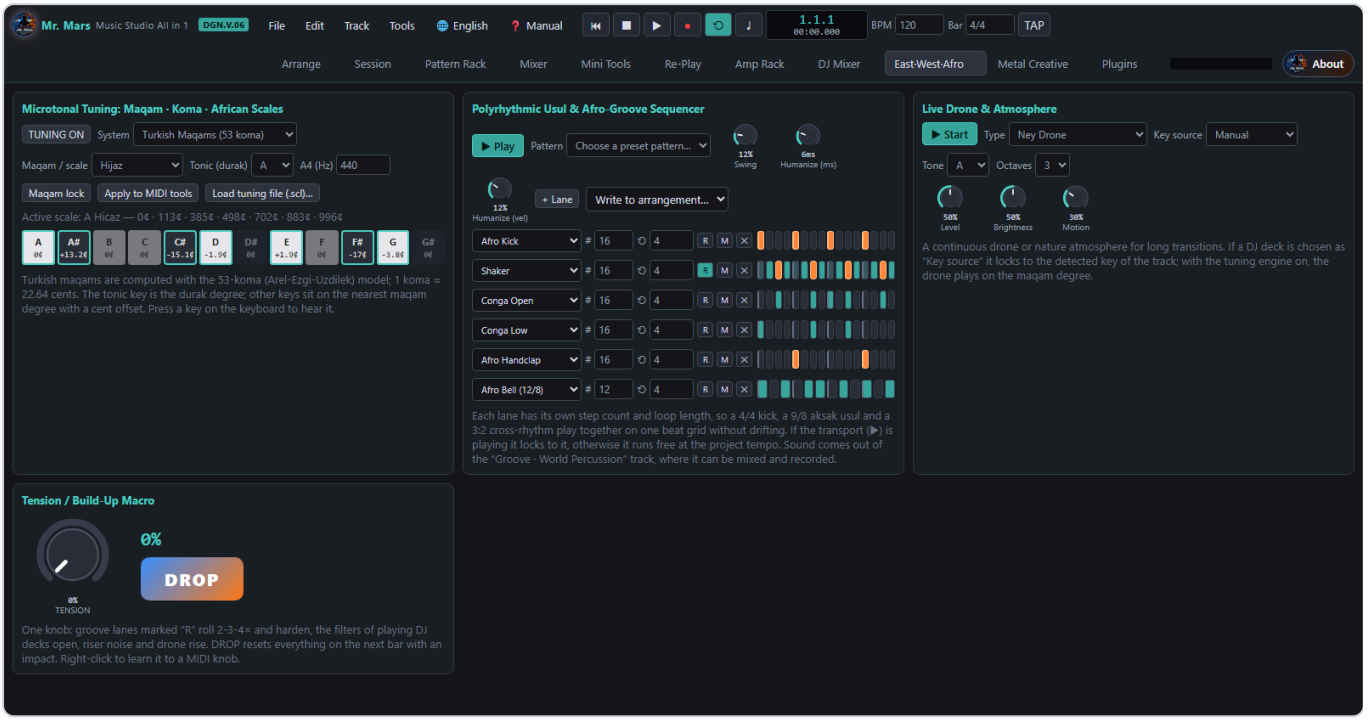
- **Jog sensitivity:** for a jog that feels too fast or too slow.
- **Tempo fader direction:** Tick Reverse if the fader runs backwards.
- **Assign a control:** touch a control on your device and choose its command.
- **Soft takeover:** a fader has no effect until it reaches the value on screen, so the sound does not jump.
- **Live signal:** shows what your device is sending.

## Computer keyboard in the DJ Mixer

| Key | Function |
|---|---|
| `Space` / `Enter` | Deck A / B play or pause |
| `1` … `8` | Deck A pads |
| `Q` … `I` | Deck B pads |
| `Z` / `X` | Deck A / B cue |
| `C` / `V` | Deck A / B sync |
| `←` / `→` | Crossfader |
| `Shift` + pad | The pad's second function |

> Not included: broadcasting, timecode vinyl, light control (DMX) and external audio plugins. Ready mappings cover common models only; other devices are mapped once with the wizard. The DJ Mixer has not been tested with every controller on real hardware.

# 12. East·West·Afro tab




Microtonal tuning, polyrhythmic sequencer, drone and Tension macro.

## Microtonal Tuning

- **TUNING ON:** every instrument plays in the selected maqam or scale.
- **System:** Turkish Maqams (53 koma), Arabic Maqams (quarter tone), African Scales or Western.
- **Maqam / scale, Tonic (durak), A4 (Hz):** the scale, its base note and the reference pitch.
- **Maqam lock:** keys outside the maqam snap to the nearest degree.
- **Apply to MIDI tools:** sets the same scale in Mini Tools.
- **Load tuning file (.scl)...:** uses your own tuning.

Turkish maqams use the 53-koma model, where one koma is about 22.64 cents. The keys show each note's offset in cents. Press a key to hear it.

## Polyrhythmic Usul & Afro-Groove Sequencer

Each lane has its own step count (#) and loop length (↻), so a 4/4 kick, a 9/8 aksak usul and a 3:2 cross-rhythm play together without drifting. Sound comes from a "Groove · World Percussion" track, where it can be mixed and recorded.

- **Pattern:** presets such as Afro House, Amapiano, Son Clave, Aksak 9/8, Curcuna 10/8 and Çiftetelli.
- **Swing, Humanize (ms), Humanize (vel).**
- **+ Lane:** adds a lane. **R:** rolls during the Tension macro. **M:** mutes the lane.
- **Write to arrangement...:** writes 1 to 16 bars.

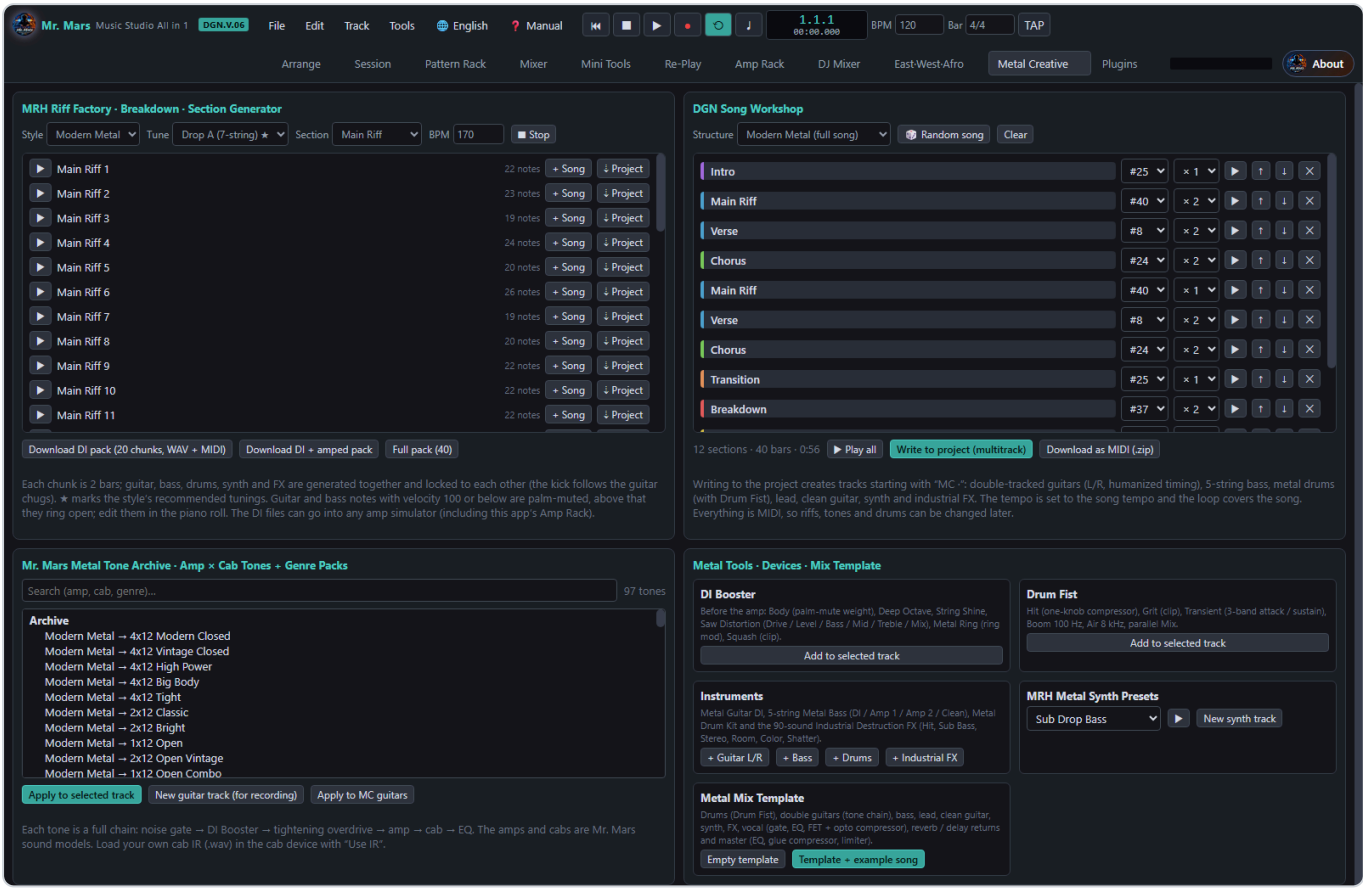
## Live Drone & Atmosphere

A continuous drone or nature sound for long transitions: Ney, Tambur / Tanpura, Analog Pad, Afro Forest, Wind, Rain or Tribal Choir. If you choose a DJ deck as "Key source", the drone locks to that track's detected key.

## Tension / Build-Up Macro

One knob raises tension: groove lanes marked R roll and harden, the filters of playing DJ decks open, and riser noise and drone rise. **DROP** resets everything on the next bar with an impact. Right-click the knob to link it to a MIDI knob.

# 13. Metal Creative tab




Riff Factory, Song Workshop, Tone Archive and Metal Tools.

## MRH Riff Factory

Choose **Style** (8), **Tune** (9, where ★ marks those recommended for the style), **Section** (8) and **BPM**. Each combination gives 40 chunks of 2 bars. Guitar, bass, drums, synth and FX are generated together, and the kick follows the guitar.

- ▶ previews a chunk. **+ Song** adds it to the Song Workshop. ⇣ **Project** writes it at the playhead.
- **Download DI pack:** a ZIP of 20 chunks with DI guitar and bass WAVs and MIDI files. "Download DI + amped pack" adds amped guitars, and "Full pack" makes 40 chunks.

## DGN Song Workshop

Choose a **Structure** and press 🎲 **Random song**, or add chunks by hand. Each row sets the chunk number and repeats, with buttons to preview, move and remove.

- **Write to project (multitrack):** creates tracks starting with "MC ·": Guitar L and R, Bass, Drums, Lead, Clean Guitar, Synth and Industrial FX. It also sets the tempo and loop.
- **Download as MIDI (.zip).**

Everything is MIDI, so riffs, tones and drums can be changed later. Guitar and bass notes with velocity 100 or below are palm-muted; higher notes ring open.

## Mr. Mars Metal Tone Archive

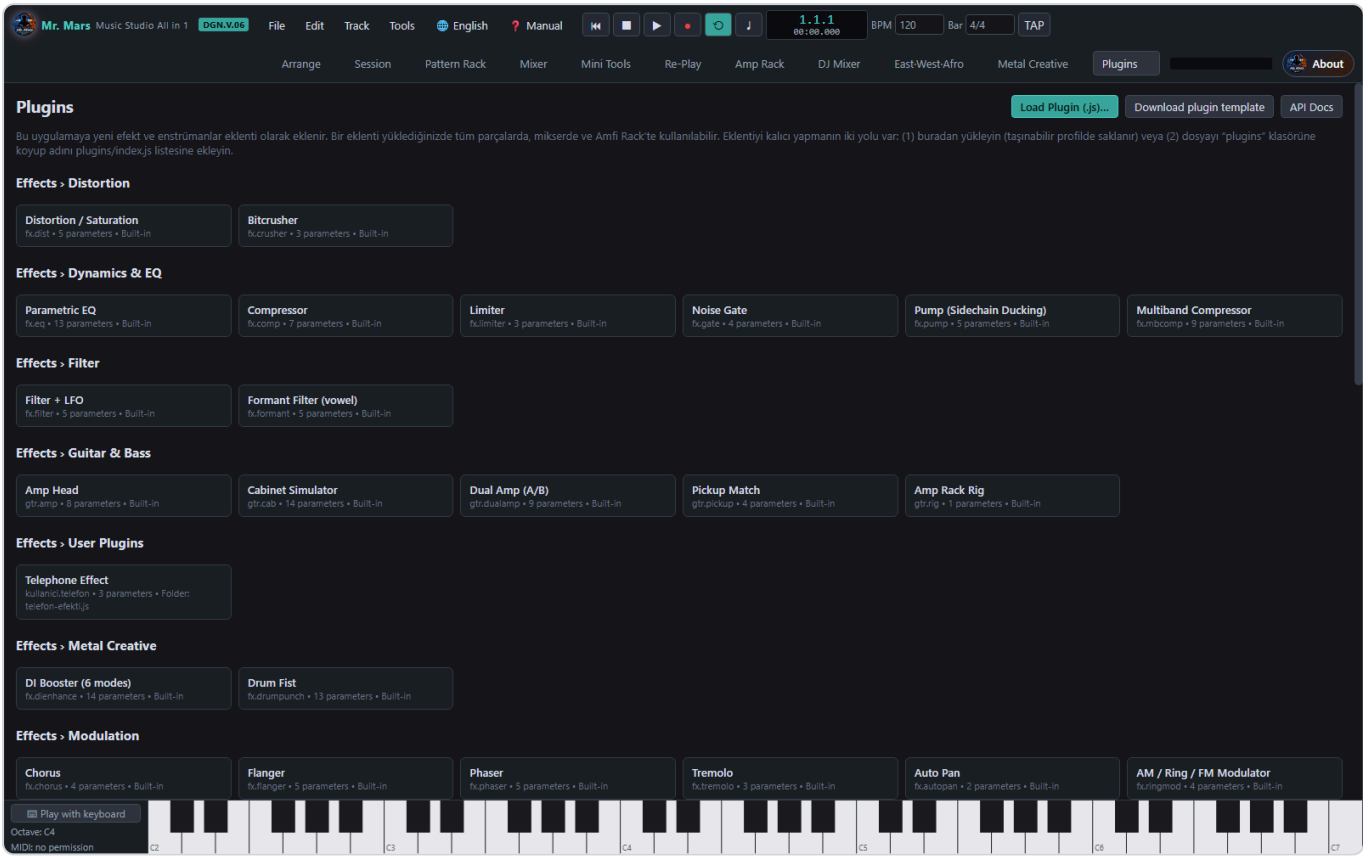
97 ready amp and cab tones, plus genre packs. Each is a full chain: noise gate, DI Booster, tightening overdrive, amp, cab and EQ. Search the list, then choose "Apply to selected track", "New guitar track" or "Apply to MC guitars".

## Metal Tools

- **DI Booster:** used before the amp. Body, Deep Octave, String Shine, Saw Distortion, Metal Ring and Squash.
- **Drum Fist:** Hit, Grit, 3-band Transient, Boom 100 Hz, Air 8 kHz and parallel Mix.
- **Instruments:** Metal Guitar DI (7/8-string), Metal Bass (5-string), Metal Drum Kit and the 90-sound Industrial Destruction FX.
- **MRH Metal Synth Presets:** also listed as "MC ›" in the Digital Mini Synth.
- **Metal Mix Template:** drums, double guitars, bass, lead, clean guitar, synth, FX, vocal chain, reverb and delay returns, and master processing. "Template + example song" also writes a song.

> Riffs are generated by rules, not recorded performances. Treat them as a starting point and edit them in the piano roll.

# 14. Plugins tab




All installed effects and instruments, grouped by type.

This tab lists every device with its parameter count and source. New effects and instruments are added as plugins. A loaded plugin can be used on every track, in the mixer and in Amp Rack.

## Adding a plugin

1. **Load Plugin (.js)...** stores it in the app's portable profile.
2. Or put the `.js` file into the `mrmars-eklentiler` folder and add its name to `mrmars-eklentiler/index.js`.

For writing your own, see **Download plugin template** and **API Docs**. Plugins register with `MrMars.register({ ... })`. The files `rotary-hoparlor.js` and `telefon-efekti.js` are examples.

> Only load plugin files you trust. A plugin is code that runs inside the app. External audio plugin formats cannot be loaded.

# 15. Connecting devices

Use Edge or Chrome on Windows and Mac. Plug the device in; there is no setup program. The browser asks for permission the first time, so choose Allow. The "MIDI:" line at the bottom left shows what the app sees.



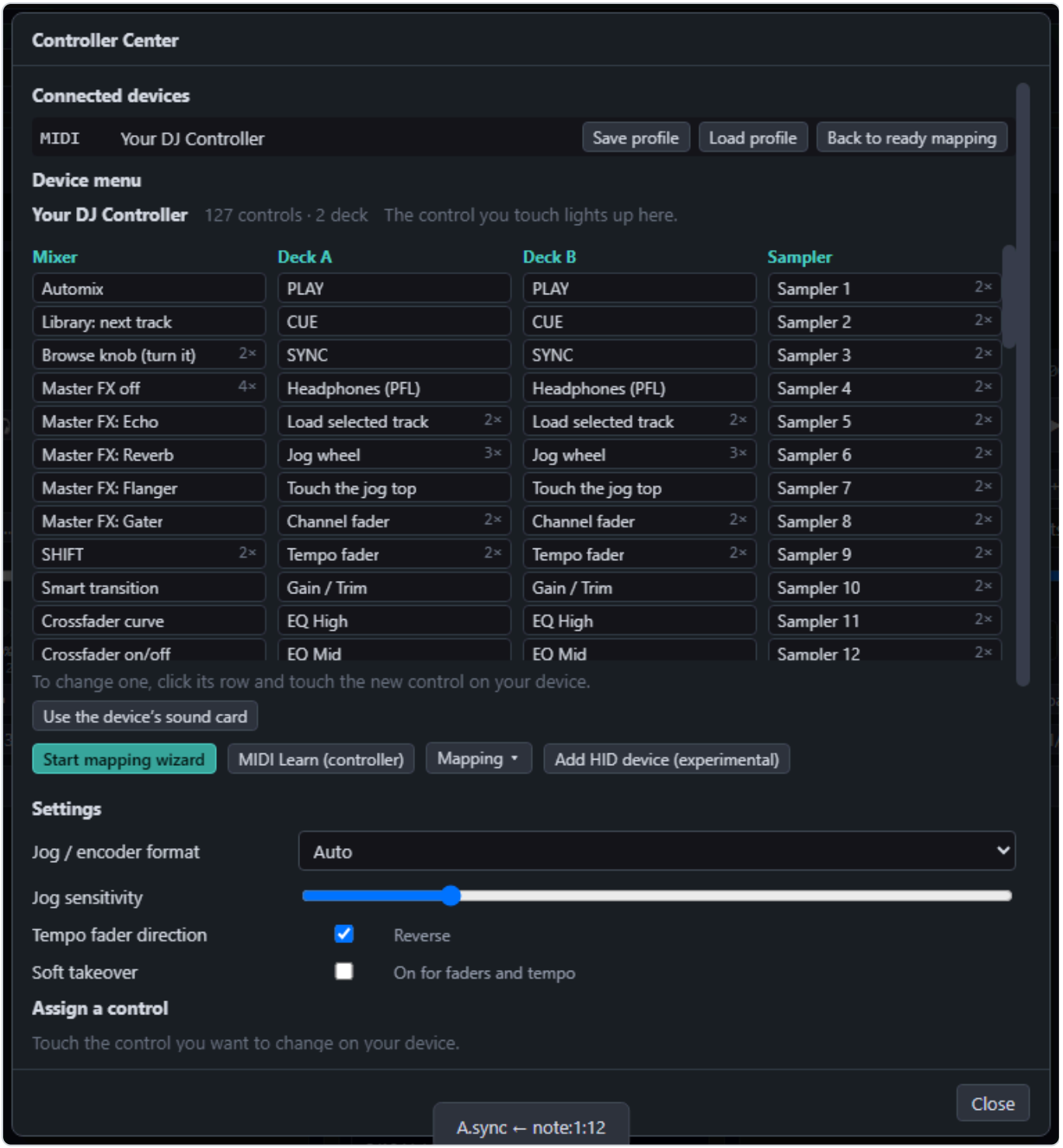

The bottom bar. This picture shows "MIDI: no permission" because no permission had been given.

## MIDI keyboard

It is recognised automatically, and notes play the selected instrument straight away.

## DJ controller

**Common models** are recognised when plugged in, and a ready mapping loads by itself. You can start playing straight away. A banner offers the **Device menu**, and a button with your device's name appears in the DJ Mixer's top bar.



The Device menu inside Controller Center lists the connected device's own controls.

The menu groups your device's controls into Mixer, decks and Sampler. The control you touch lights up. To change one, click its row and touch the new control on your device.

**Other models** are mapped once:

1. Plug it in. A banner offers **Map**, **Not now** and **Don't ask again**.
2. **Map** opens the wizard. Move each control when asked, and skip what your device lacks.
3. The mapping is saved as a profile for that device, and loads by itself next time.

**To change any control**, open **DJ Mixer > Controller Center**, go to "Assign a control", touch the control on your device and choose a command. "Back to ready mapping" undoes your changes.

**Jog wheels:** touch the top and turn to scratch; the sound follows your hand, forwards and backwards. When you let go, the track carries on from there. Turn the edge to nudge the tempo during play. This needs Vinyl on in the DJ Mixer.

**SHIFT:** on recognised devices with a SHIFT layer, the Device menu lists what each SHIFT combination does. You can change any of them there.

**Sound card in the controller:** if your controller has its own sound card, press "Use the device's sound card" in the Device menu. When it has four output channels, headphones become a separate cue output.

> Ready mappings have not been tried on every device. If something does not respond, assign it yourself or run the wizard.

## Electric guitar or bass

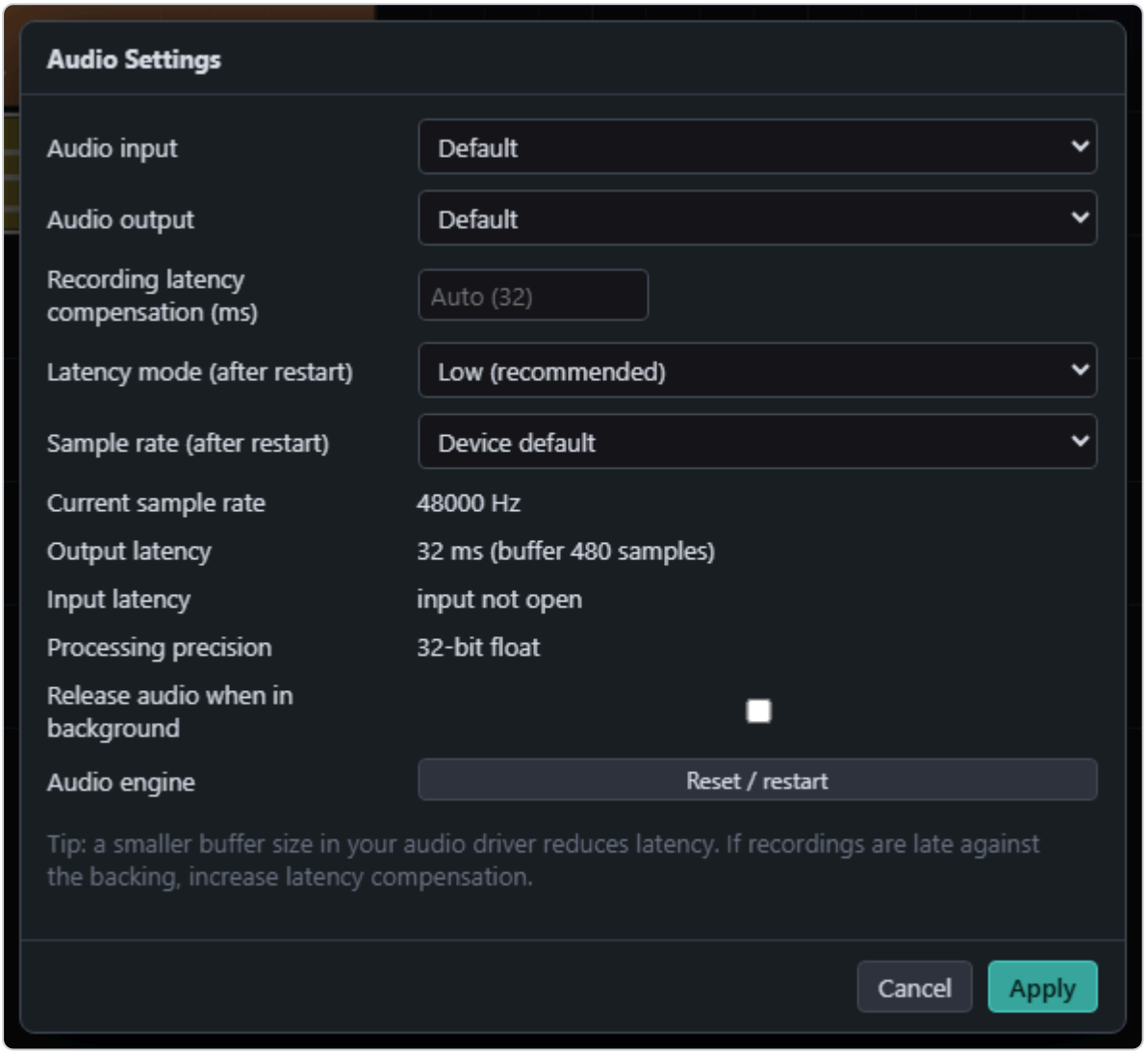
1. A guitar does not connect by MIDI. Plug it into the Hi-Z (instrument) input of an audio interface.
2. Connect the interface by USB.
3. Choose it as input and output under **Tools > Audio settings**.
4. Open **Amp Rack**, press ⏻ and allow the microphone.
5. Set the level on the interface's gain knob so that it does not clip.

## Windows and Mac differences

| | **Windows** | **Mac** |
|---|---|---|
| Driver | USB-MIDI usually works without one. A controller's built-in sound card may need the maker's driver. | Usually none needed. |
| Other programs | A MIDI port usually serves one program, so close other DJ or music programs first. | Ports can be shared. |
| Microphone permission | Settings > Privacy & security > Microphone | System Settings > Privacy & Security > Microphone |
| Browser | Edge or Chrome | Edge or Chrome. Safari has no MIDI. |

# 16. Audio, MIDI and export settings

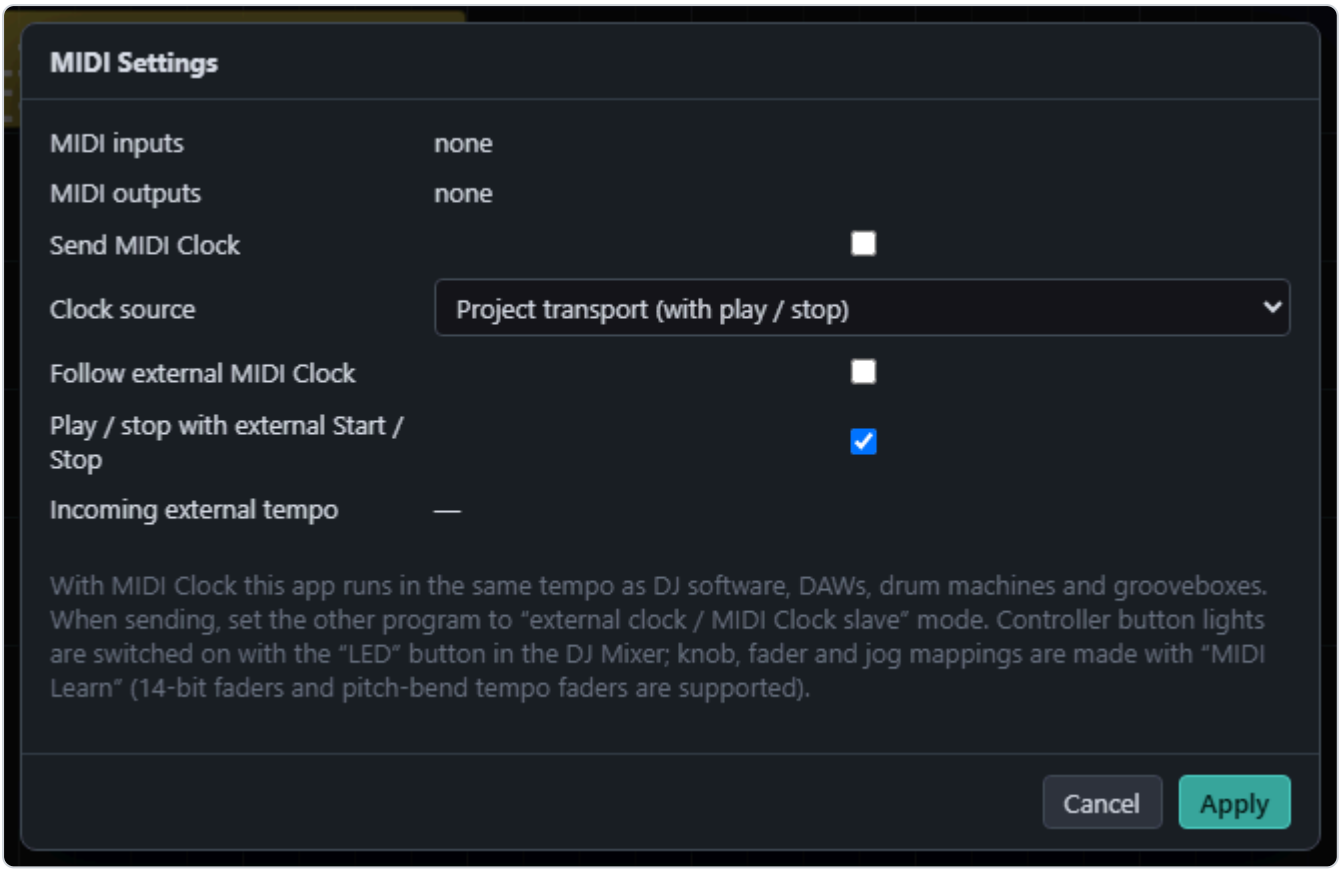
## Audio settings




- **Audio input and output:** your interface or sound device. Your own device names appear here.
- **Recording latency compensation:** if recordings land late against the backing, raise this value. Leave it empty for automatic.
- **Latency mode:** Lowest, Low, Balanced or Safe. Applies after restart.
- **Sample rate:** device default, or 44.1 to 96 kHz. Applies after restart.
- **Release audio when in background:** frees the sound device when the app is not in use.
- **Reset / restart:** restarts the audio engine.

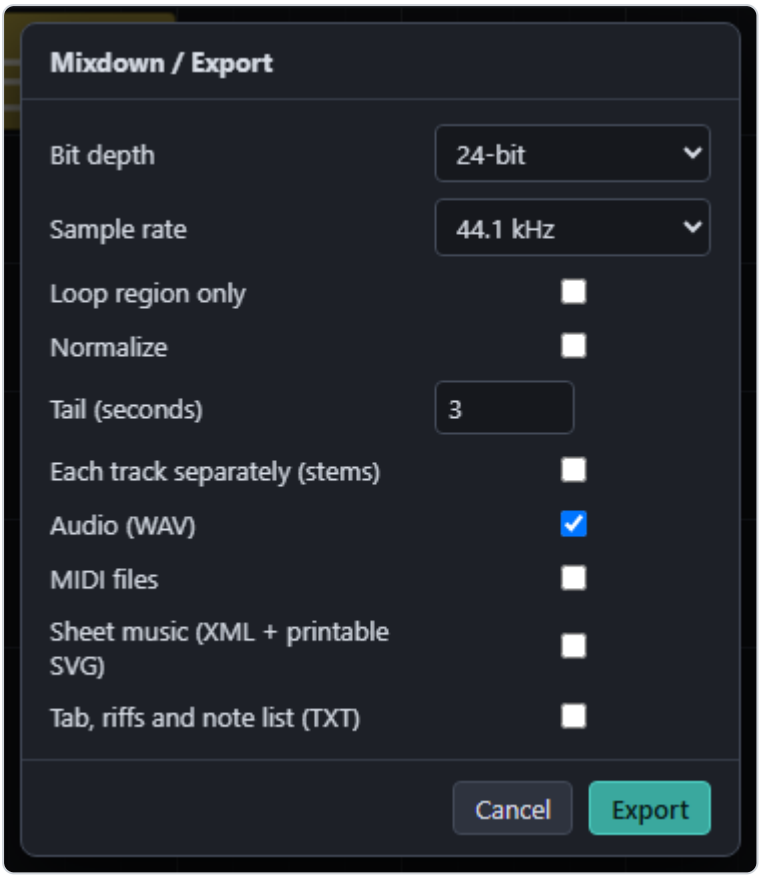
If the sound crackles, choose Safe and run fewer amp chains at once. Browsers use the normal system audio path, not special low-latency drivers.

# MIDI settings



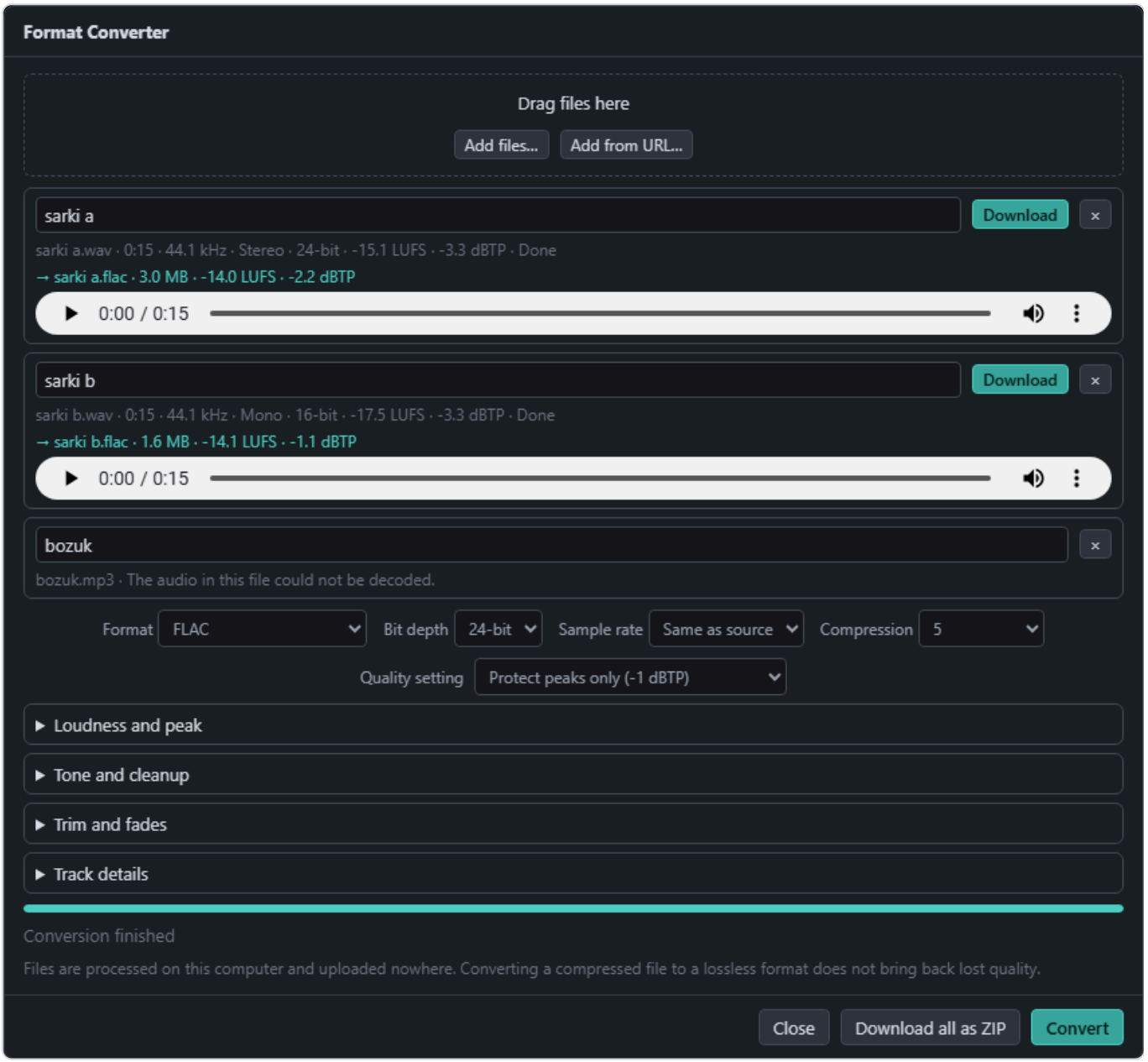

- **MIDI inputs and outputs:** the connected ports.
- **Send MIDI Clock:** other programs and machines follow this app's tempo. Set the other device to "external clock".
- **Clock source:** the project transport, or the playing DJ deck.
- **Follow external MIDI Clock:** this app follows another device.
- **Play / stop with external Start / Stop.**

# Mixdown / Export




- **Bit depth:** 16-bit, 24-bit or 32-bit float.
- **Sample rate.**
- **Loop region only:** exports only the marked region.
- **Normalize:** raises the level to full scale.
- **Tail (seconds):** extra time for reverbs and delays to fade.
- **Each track separately (stems):** one WAV per track.
- **Audio (WAV), MIDI files.**
- **Sheet music (XML + printable SVG).**
- **Tab, riffs and note list (TXT).**

# Format Converter




Many files at once, with their measurements before and after.

Open it from **Tools** ▸ **Format Converter** or from the Send to Phone window. All the work happens on this computer; nothing is uploaded.

- **Add files:** drag them in, choose them, or add them from a web address if the site allows it. Audio inside MP4 and WebM videos also works.
- **Output:** WAV (16, 24 or 32-bit float), AIFF, FLAC with compression level, Apple Lossless, MP3 (64 to 320 kbps), AAC, Opus or MP4. Choose the sample rate, up to 192 kHz for lossless formats.
- **Results:** each file shows its loudness and true peak before and after. Listen, press **Download**, send it **To phone** if a phone is connected, or press **Download all as ZIP**.

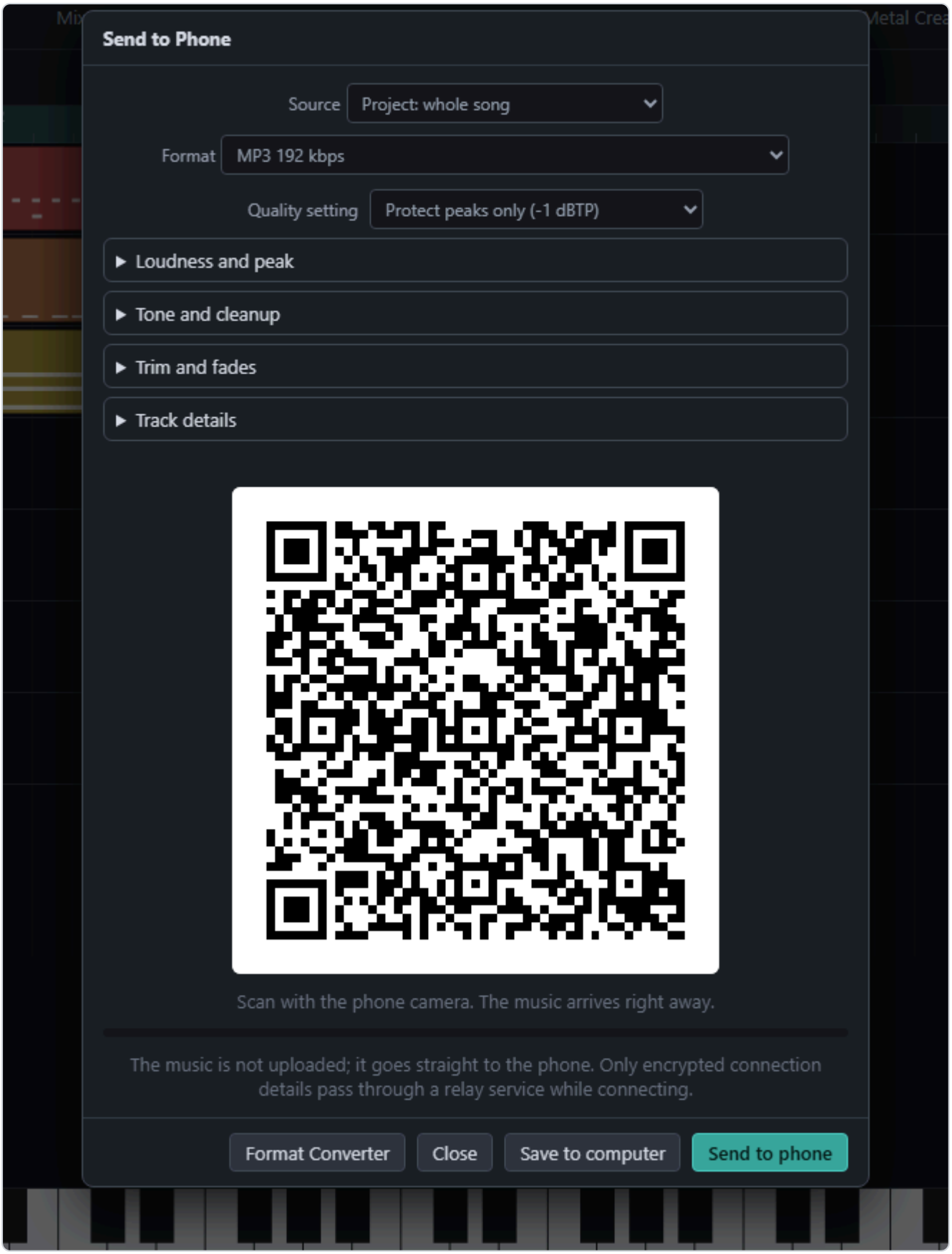
## Quality settings

The same panel appears in Send to Phone, Mixdown and Format Converter, and the settings are shared. Choose a ready setting or open the sections:

| Section | Settings |
|---|---|
| Loudness and peak | Method (do not change, lower if over, normalize to peak, LUFS target), peak ceiling, true peak limiter, volume. |
| Tone and cleanup | High-pass, bass, treble, stereo width, compressor, dither for 16-bit. |
| Trim and fades | Start and end points, fade in, fade out, silence removal, channels. |
| Track details | Title, artist, album, year, genre, track number, ISRC and cover. WAV and Opus files have no cover; M4A files have no ISRC. |

These settings do not replace mixing and mastering, and they cannot restore quality already lost to compression. With **Off** and no other changes, a WAV or AIFF file passes through bit for bit.

# Send to Phone




Scan the code with the phone and the music arrives.

The **Send to Phone** button is in the top bar, so it works from every tab.

1. Choose the **Source**: the latest DJ mix (up to 30 minutes), the whole project, or the loop region.
2. Choose the **Format** (see the table below).
3. Scan the QR code with the phone's camera. The music arrives right away. Or press **Save to computer**.
4. Press **Save** or **Share** on the phone.

On the phone, choose another format from the list and press **Get**. On the computer, **Send to phone** sends again without scanning.

| Format | Use |
|---|---|
| MP3 192 / 320 kbps | Quick listening and sharing. |
| AAC M4A | Apple devices. Uses the highest rate your system allows, up to 256 kbps; the status line shows the actual rate. |
| MP4 | Music with a moving waveform cover. |
| WAV 24-bit 44.1 kHz | Recommended for distribution services. |
| WAV 24-bit 48 / 96 kHz | Higher-resolution masters. |
| WAV 16-bit 44.1 kHz | CD standard, with dither. |
| FLAC 16 / 24-bit | Same audio as WAV in a smaller file. |
| Apple Lossless M4A | Lossless for Apple players. The file is about the size of a WAV. |

For distribution, use the project or a **Record Set** WAV recording as the source. The "latest mix" DJ source comes from a 128 kbps compressed buffer, so a lossless file made from it does not gain quality, and the window warns you. A recording cannot be raised to a higher sample rate. **Limit peaks to -1 dBFS** lowers the whole level only if the peaks are above that point. Check your distributor's current rules before uploading.

- Both devices need internet to connect, and the window must stay open on the computer until the phone connects.
- The music goes directly between the devices and is not uploaded. Only encrypted connection details pass through public relay services.
- The same Wi-Fi works best. On mobile data the direct connection can fail on some networks.
- Anyone who scans the code while the window is open can receive the music.
- Keep the phone's screen on during the transfer, and save the file before you leave the page.

# 17. Mac, phones and tablets

## Mac

Double-click `Mr Mars Music Studio.command`. Chrome or Edge is required. If macOS blocks it the first time, right-click it and choose Open. If it says permission denied, run this once in Terminal from the app folder:

```
chmod +x "Mr Mars Music Studio.command"
```

## Android, iPhone and iPad




The app on a narrow screen.

The app must be opened from its web address, once, while online. Then add it to the home screen:

- **Android:** Tools > "Add to home screen".
- **iPhone or iPad:** Safari > Share > "Add to Home Screen".

It then opens offline too. Projects stay in that device's browser, so export a `.mrmars` file to move them.

### Touch controls

- **Press and drag** moves controls.
- **Long press** acts as right-click.
- **Double tap** resets a knob.
- **☰ and** 🎹 open the panel and keyboard on narrow screens.
- **SHIFT** in the DJ Mixer gives pads their second function.

### Limits

- MIDI works on Android in Chrome with USB OTG. No browser on iPhone or iPad has MIDI.
- Phones have less CPU and memory. Several amp chains, stem separation and long tracks can be too heavy.
- The iPhone silent switch can mute the sound.
- Some devices cannot decode formats such as FLAC or OGG.
- A separate headphone output is rarely available, so use "Split" mode.

> Mac, Android and iOS support has not been tested on every real device.

# 18. Keyboard shortcuts




Tools > Keyboard shortcuts shows this list in the app.

| Key | Function |
|---|---|
| `Space` | Play or stop |
| `R` , `L` , `M` | Record, loop, metronome |
| `Home` | Back to the start |
| `S` , `Del` | Split at playhead, delete selected clips |
| `Ctrl` + `Z` / `Y` | Undo, redo |
| `Ctrl` + `C` / `V` / `D` | Copy, paste, duplicate |
| `Ctrl` + `S` , `Ctrl` + `A` | Save, select all clips |
| `Ctrl` + mouse wheel | Horizontal zoom |

| Key | Function |
| --- | --- |
| `Alt` + drag, `Shift` + drag | Copy a clip while moving; ignore the grid |
| Drag in the ruler | Select the loop region |
| `Q` in the piano roll | Quantize |

## Play with keyboard

Press "⌨ Play with keyboard" at the bottom left to play notes with the computer keyboard:

- `A W S E D F T G Y H U J K`: notes
- `Z` / `X`: octave down or up
- `C` / `V`: velocity down or up

The DJ Mixer has its own keys. See Controller Center.

# 19. Files, backup and moving

## Moving to another computer

Copy the whole "Mr Mars Music Studio All in 1" folder to a USB drive. Projects, recordings, settings and installed plugins live in the `veri` folder inside it. This applies when you start the app with its launcher.

## Backup and sharing

**File > Export project file (.mrmars)** creates one complete file with the project and its audio. This is the safest backup. Keep copies outside the app folder.

## Folder layout

| Folder | Contents |
|---|---|
| `mrmars-cekirdek` | Audio engine, project, automation, DSP, tuning, MIDI, translations |
| `mrh-cihazlar` | Instruments, effects, amps, cabs, pedals, metal devices |
| `dgn-arayuz` | Tabs, windows, DJ, Metal Creative, plugin manager |
| `dgn-tema` | Interface theme |
| `mrmars-gorseller` | Logo and icons |
| `mrmars-eklentiler` | Plugins you add |
| `veri` | Your projects and settings |

## License

The project's own code is under the MIT License (LICENSE file). Third-party components keep their own licenses. The MP3 encoder is LAME / lamejs under LGPL-3.0 (lame.sourceforge.net); see THIRD-PARTY-NOTICES.txt.

# 20. Troubleshooting and known limits

| Problem | What to do |
|---|---|
| "MIDI: no permission" | Allow MIDI in the site permissions (the padlock in the address bar), then reload. |
| "MIDI: no device" | Try another USB port or cable, close other music programs, and reload. |
| "MIDI: not supported" | Switch to Edge or Chrome. |
| No signal in the mapping wizard | Some devices use HID rather than MIDI. Try "Add HID device (experimental)" in Controller Center. Some models first need the maker's driver or switching to a MIDI mode. |
| No guitar sound | Check the input in Audio settings, the microphone permission and the interface gain. Check that ⏻ Start Rack is on. |
| Crackling sound | Choose Safe under Audio settings > Latency mode and restart. Run fewer amp chains at once. |
| Recordings are late | Raise "Recording latency compensation" in Audio settings. |
| Interface in the wrong language | Use the 🌐 button beside the menus. |

## Known limits

- The app uses the browser's audio engine. External audio plugins and special audio drivers are not used; it is extended with its own plugin system.
- Stem separation and audio-to-notes use signal processing, so results are approximate.
- Metal Creative riffs are rule-based, not recorded performances.
- There is no true sidechain. The Pump effect does tempo-locked ducking.
- There is no automation on the master channel or the Amp Rack.

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